Post-Quantum Cryptography

Johannes Buchmann and Nina Bindel

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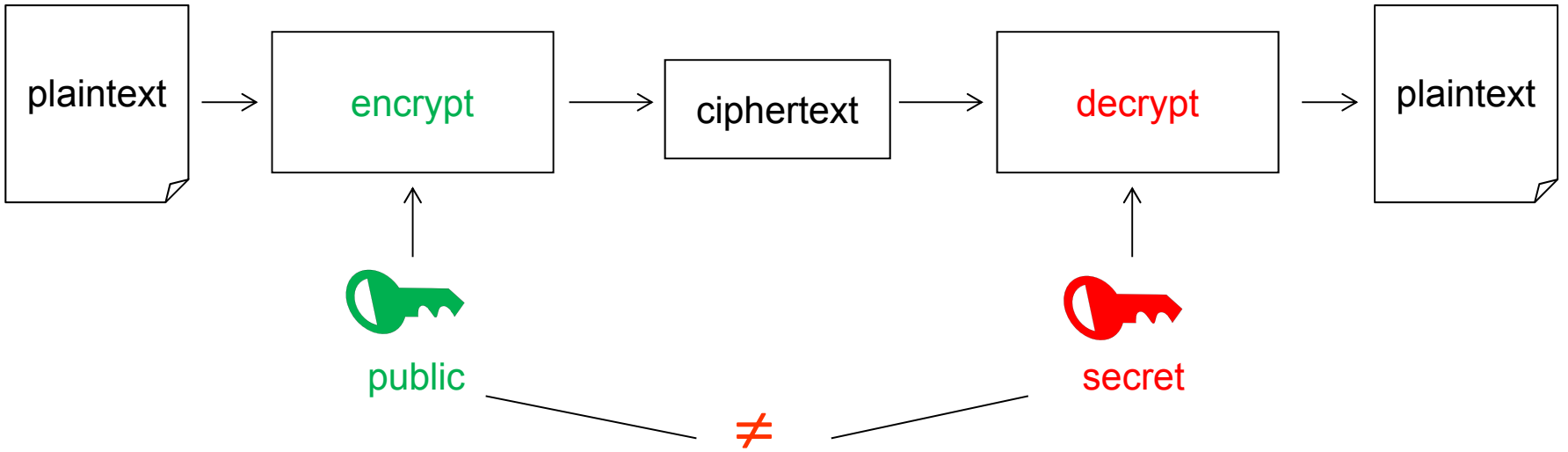
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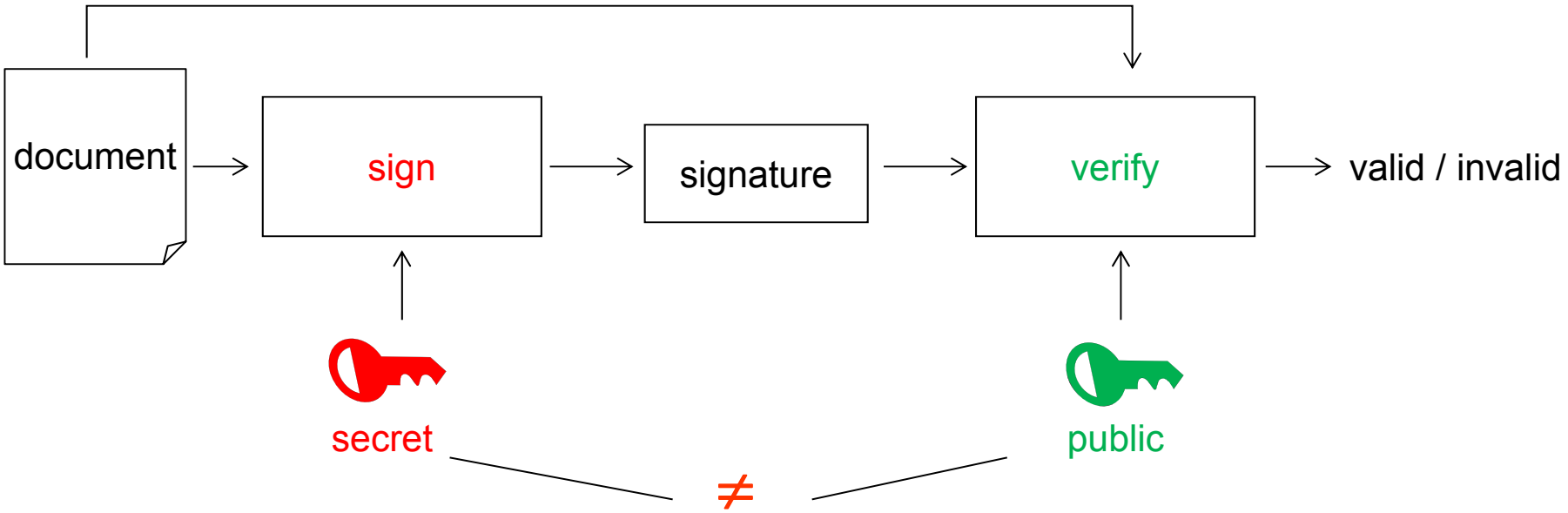
Public-key cryptography

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Public-key encryption



Digital signatures



IT-security requires public-key cryptography

Communication on the Internet

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Protected by TLS

Public-key encryption
Digital signatures



Billions of connections daily!

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Software downloads





Authenticity
protected by
digital signatures







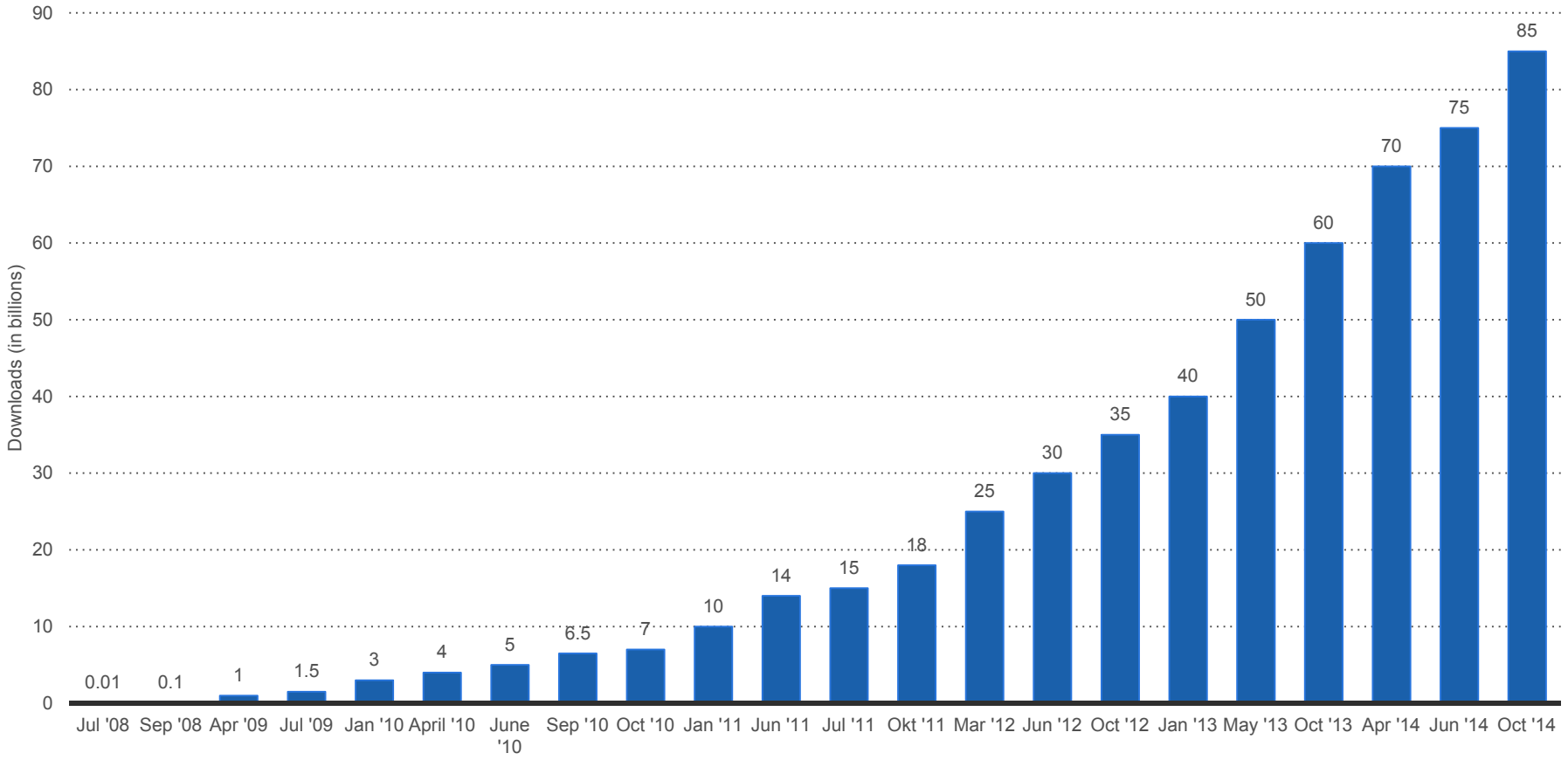




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Number of worldwide downloads from Apple App Store July 2008 - October 2014 (in billions)



Source: Apple

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Current public-key cryptography

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“Generic” RSA

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Public key: finite Group G, exponent e, $\gcd(\mathrm{e}, |\mathrm{G}|) = 1$

Secret key: $|\mathrm{G}|$

Allows to compute: $\sqrt[\mathrm{e}]{\mathrm{g}} = \mathrm{g}^{\mathrm{e}^{-1} \bmod |\mathrm{G}|}, \mathrm{g} \in \mathrm{G}$

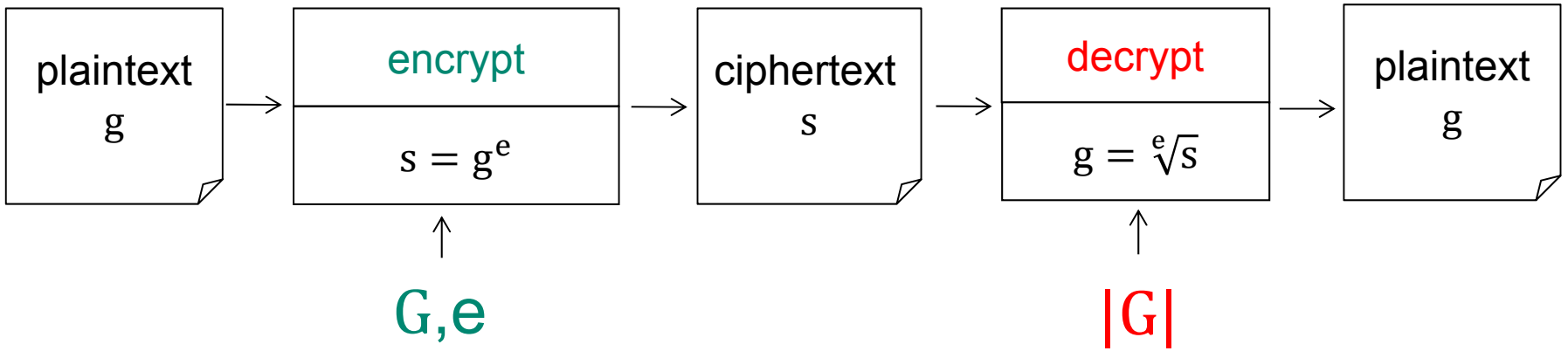
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“Generic” RSA encryption

Public key: finite Group G, exponent e, $\gcd(e, |G|) = 1$

Secret key: $|G|$

Allows to compute: $\sqrt[e]{g} = g^{e^{-1} \bmod |G|}, g \in G$



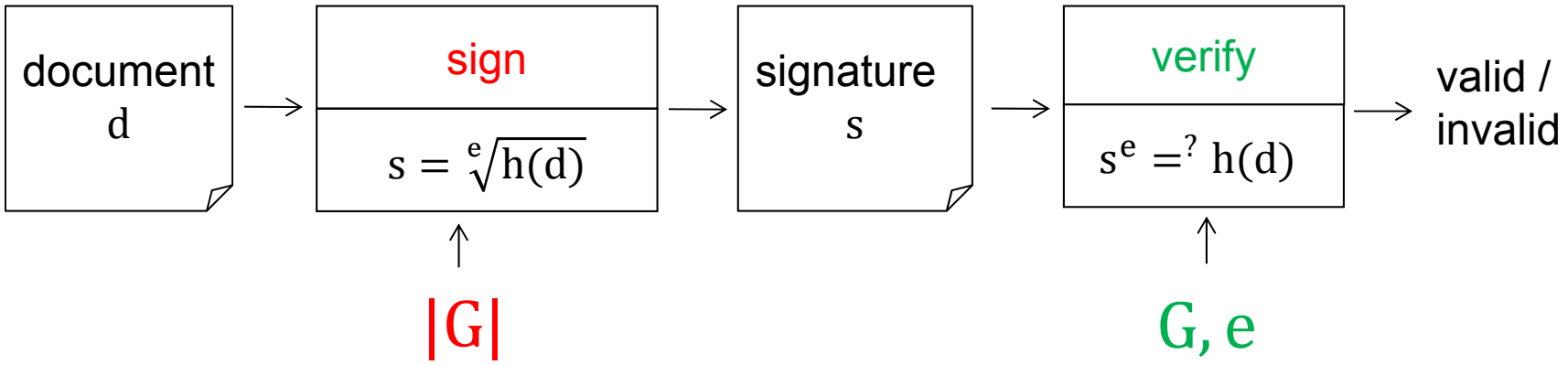
“Generic” RSA signature

Public key: finite Group G, exponent e, $\gcd(e, |G|) = 1$

Secret key: $|G|$

Allows to compute: $\sqrt[e]{g} = g^{e^{-1} \bmod |G|}, g \in G$

Hash function $h: \{0,1\}^* \rightarrow G$



RSA: How to keep $|G|$ secret?

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Set up: p, q primes, $n = pq$, $G = (\mathbb{Z}/n\mathbb{Z})^*$, e, $\gcd(e, |G|) = 1$

$|G| = (p-1)(q-1)$,

Public key: (n, e)

Secret key: $d = e^{-1} \bmod (p-1)(q-1)$

Security relies on hardness of integer factorization

Only known method to keep $|G|$ secret

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Factorization complexity

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$$L_n[u,v] = e^{v(\log n)^u (\log\log n)^{(1-u)}}$$

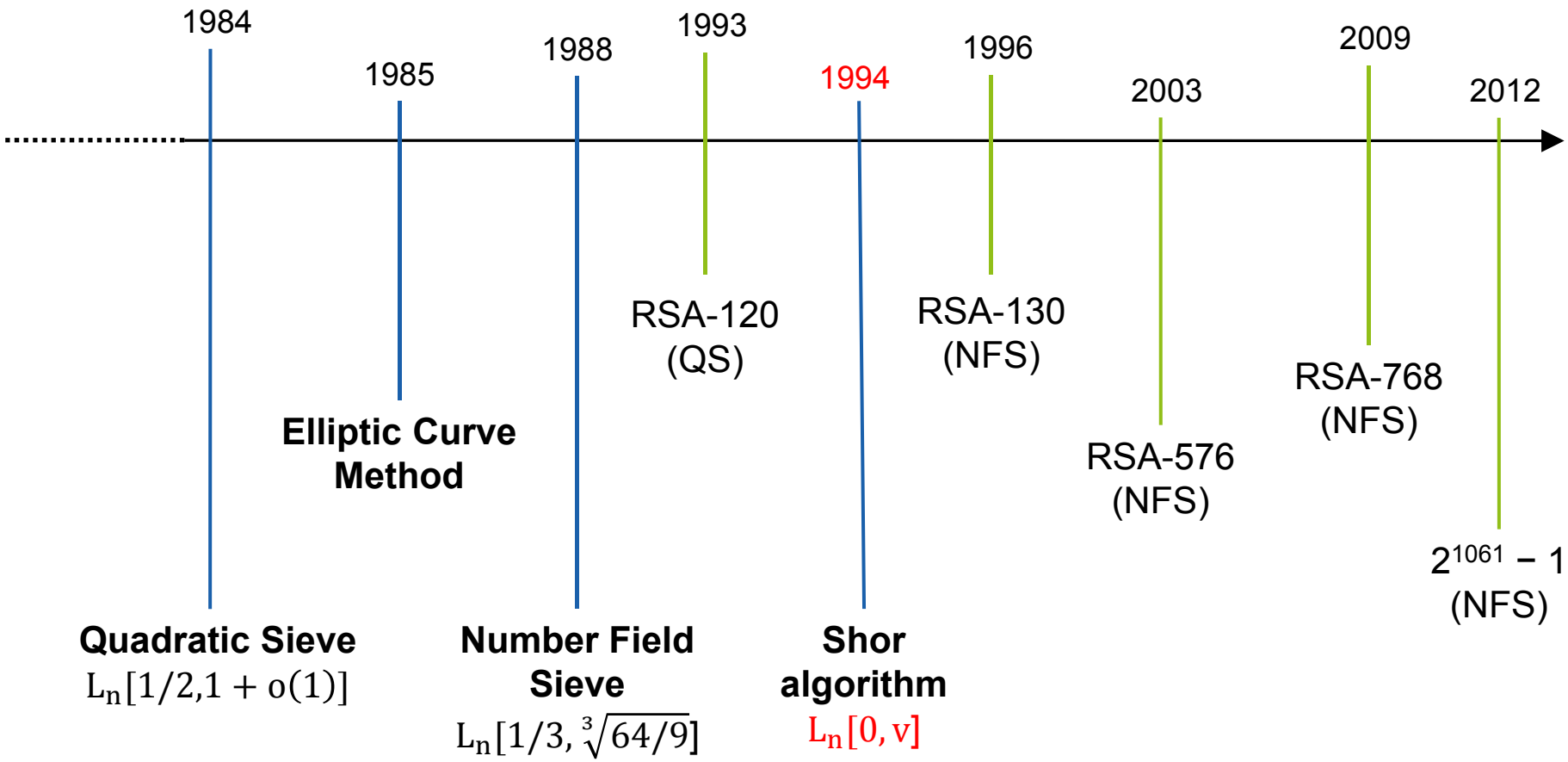
$L_n[0,v] = (\log n)^v$ polynomial

$L_n[1,v] = (e^{\log n})^v$ exponential

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Factorization progress

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ElGamal encryption and signatures

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Rely on the **Discrete Logarithm** Problem:

Given: Group $G = \langle g \rangle$, $h \in G$

Find: $x \in \mathbb{N}$ with $h = g^x$

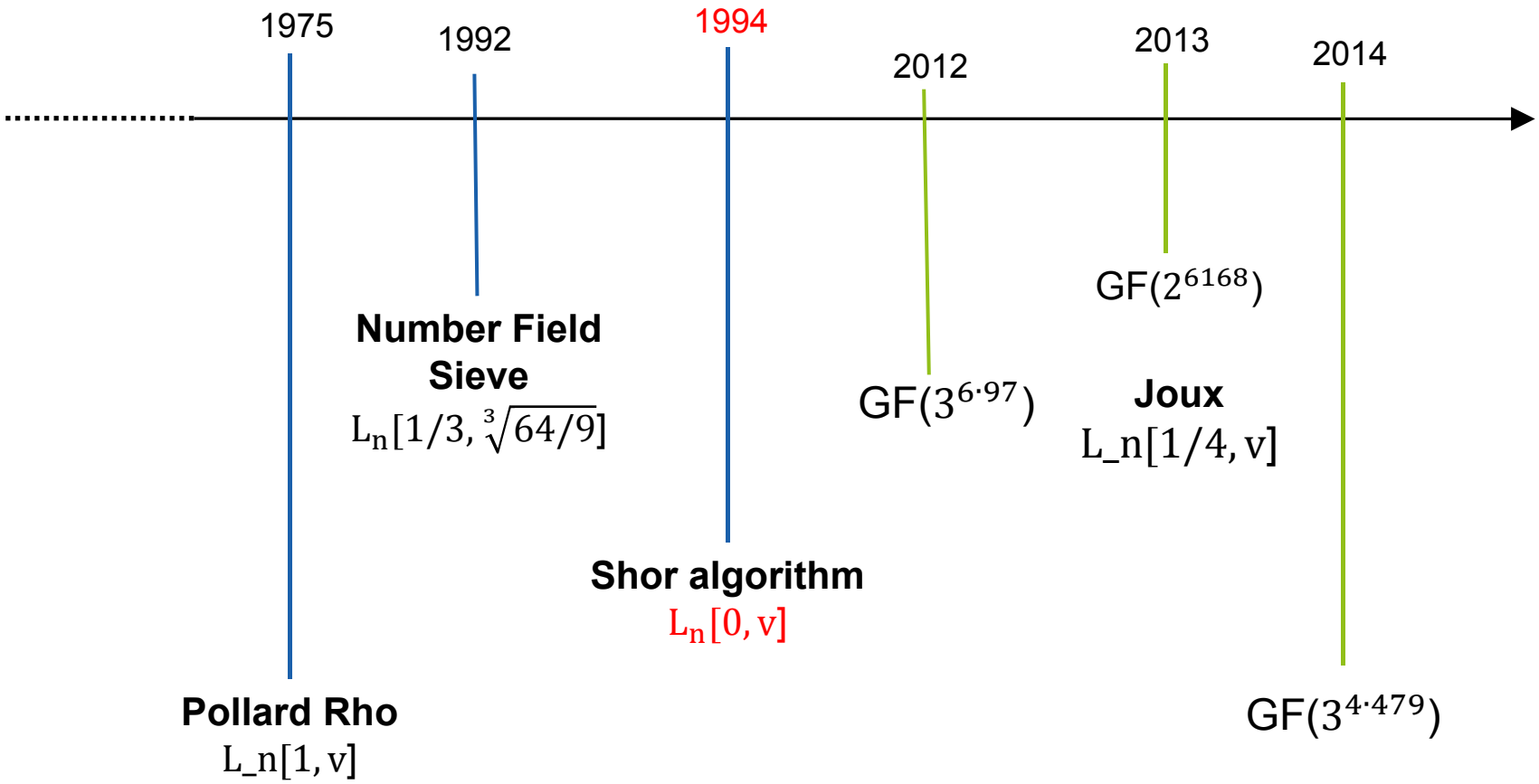
Choices for G: $GF(p^n)^*$

Group of points of elliptic curves over $GF(p^n)$

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Algorithms for solving $GF(p^n)^*$-DL

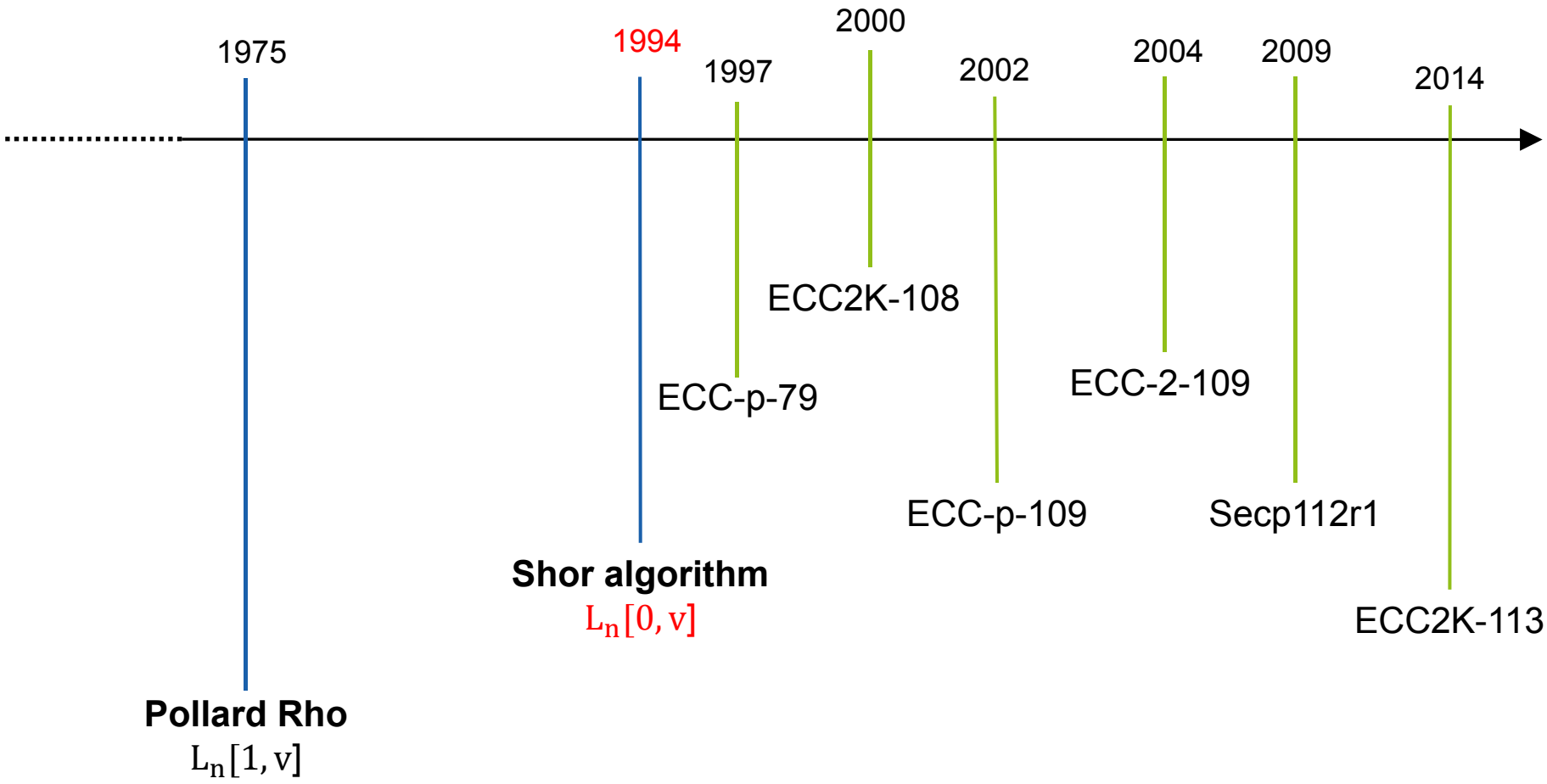
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Algorithms for solving EC-DL

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The quantum computer threat

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Shor's algorithm 1997

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Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*

Peter W. Shor†

RSA and ElGamal insecure

A digital computer is g
device; that is, it is believ
an increase in computatio
true when quantum mechanics is taken into consideration. This paper considers factoring integers and finding discrete logarithms, two problems which are generally thought to be hard on a classical computer and which have been used as the basis of several proposed cryptosystems. Efficient randomized algorithms are given for these two problems on a hypothetical quantum computer. These algorithms take a number of steps polynomial in the input size, e.g., the number of digits of the integer to be factored.

Keywords: algorithmic number theory, prime factorization, discrete logarithms, Church's thesis, quantum computers, foundations of quantum mechanics, spin systems, Fourier transforms

AMS subject classifications: 81P10, 11Y05, 68Q10, 03D10

Quantum computer realistic?

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MA Stat

NSA seeks to build quantum computer that could crack most types of encryption

By Steven Rich and Barton Gellman, Published: January 2 E-mail the writers

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Quantum computer realistic

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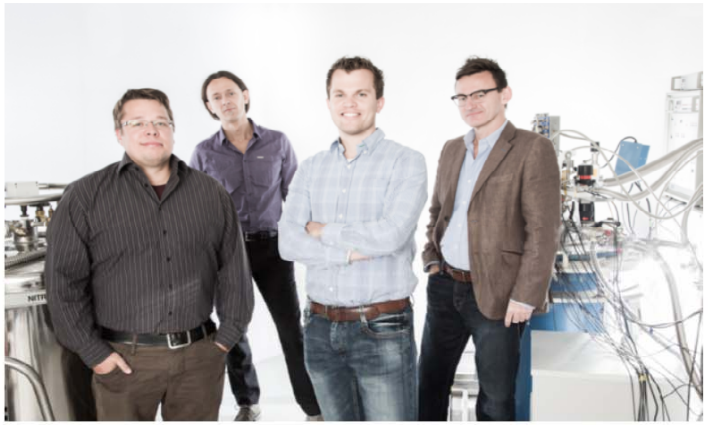
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Researchers use silicon to push quantum computing toward reality



Researchers at the University of New South Wales are pushing forward the possibility of developing a true quantum computer. From left, Juha Muhonen, Andrea Morello, Menno Veldhorst and Andrew Dzurak have been researching ways to use silicon to develop quantum bits.

Credit: University of New South Wales

New tech could let quantum machines tackle huge problems



By Sharon Gaudin FOLLOW

Computerworld | Oct 23, 2014 9:27 AM PT

Researchers in Australia have developed silicon-wrapped quantum technology

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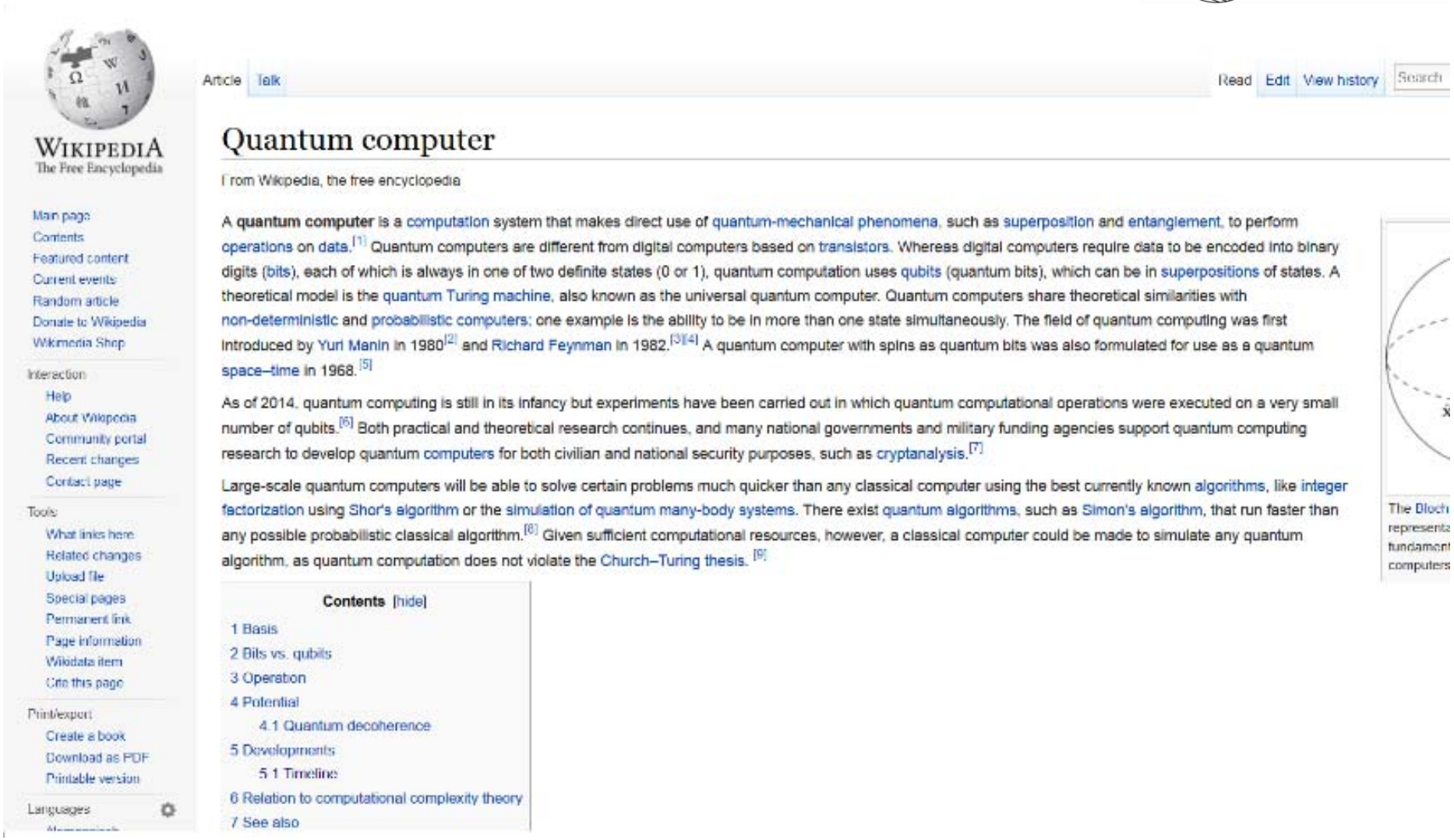
Money talks, and that's all quantum maker D-Wave has to say

ia, Santa

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s to perform

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Quantum computer

From Wikipedia, the free encyclopedia

A **quantum computer** is a computation system that makes direct use of quantum-mechanical phenomena, such as superposition and entanglement, to perform operations on data.[1] Quantum computers are different from digital computers based on transistors. Whereas digital computers require data to be encoded into binary digits (bits), each of which is always in one of two definite states (0 or 1), quantum computation uses qubits (quantum bits), which can be in superpositions of states. A theoretical model is the quantum Turing machine, also known as the universal quantum computer. Quantum computers share theoretical similarities with non-deterministic and probabilistic computers; one example is the ability to be in more than one state simultaneously. The field of quantum computing was first introduced by Yuri Manin in 1980[2] and Richard Feynman in 1982.[3][4] A quantum computer with spins as quantum bits was also formulated for use as a quantum space–time in 1968.[5]

As of 2014, quantum computing is still in its infancy but experiments have been carried out in which quantum computational operations were executed on a very small number of qubits.[6] Both practical and theoretical research continues, and many national governments and military funding agencies support quantum computing research to develop quantum computers for both civilian and national security purposes, such as cryptanalysis.[7]

Large-scale quantum computers will be able to solve certain problems much quicker than any classical computer using the best currently known algorithms, like integer factorization using Shor's algorithm or the simulation of quantum many-body systems. There exist quantum algorithms, such as Simon's algorithm, that run faster than any possible probabilistic classical algorithm.[8] Given sufficient computational resources, however, a classical computer could be made to simulate any quantum algorithm, as quantum computation does not violate the Church–Turing thesis.[9]

Contents [hide]

1 Basis
2 Bits vs. qubits
3 Operation
4 Potential
4.1 Quantum decoherence
5 Developments
5.1 Timeline
6 Relation to computational complexity theory
7 See also

Post-quantum cryptography

Performance requirements

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Secure until	Security level	RSA modulus/finite field size	Elliptic curve
2015	80	1248	160
2025	96	1776	192
2030	112	2493	224
2040	128	3248	256

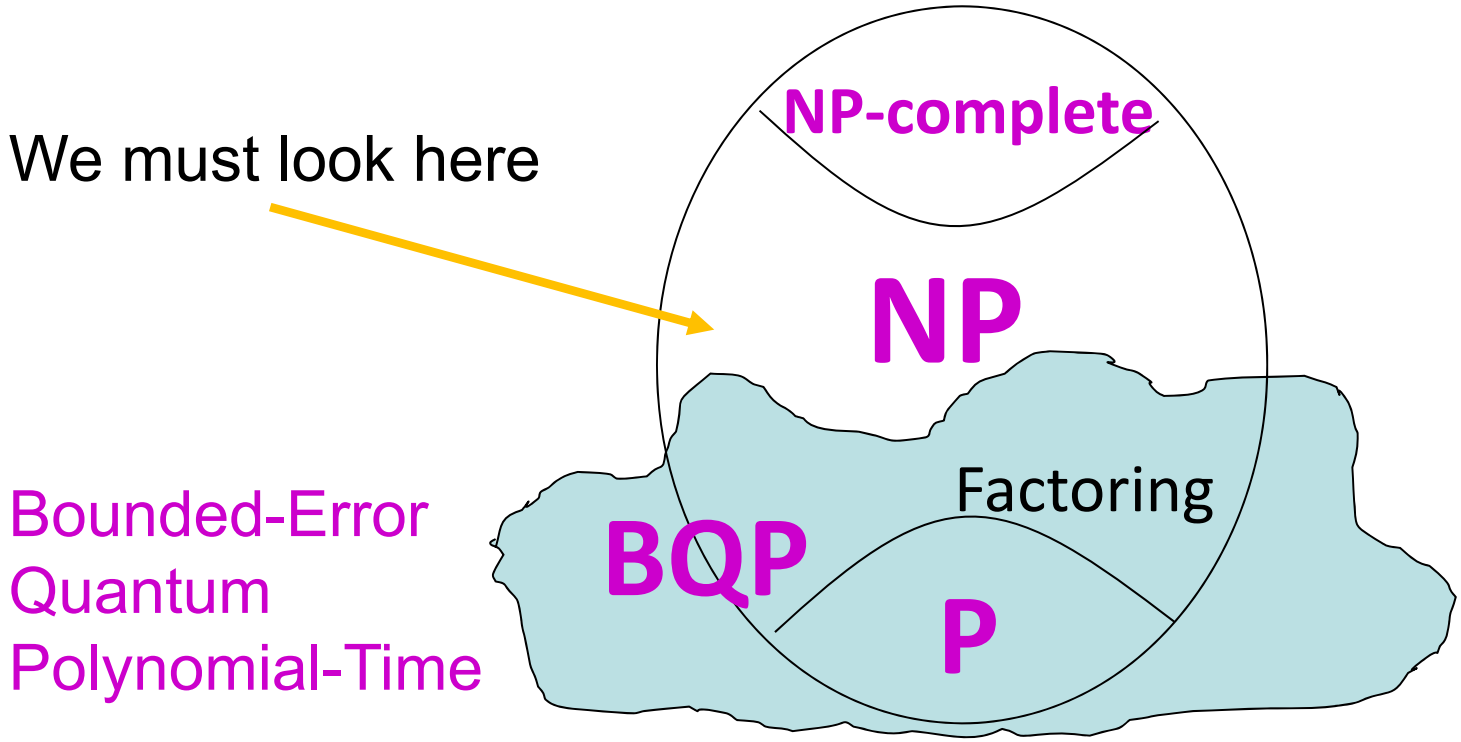
Ecrypt recommendations

- Space for keys and signatures: a few kilobytes
- Small ciphertext expansion
- Times: milliseconds

Post-quantum problems?

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No provably quantum resistant problems

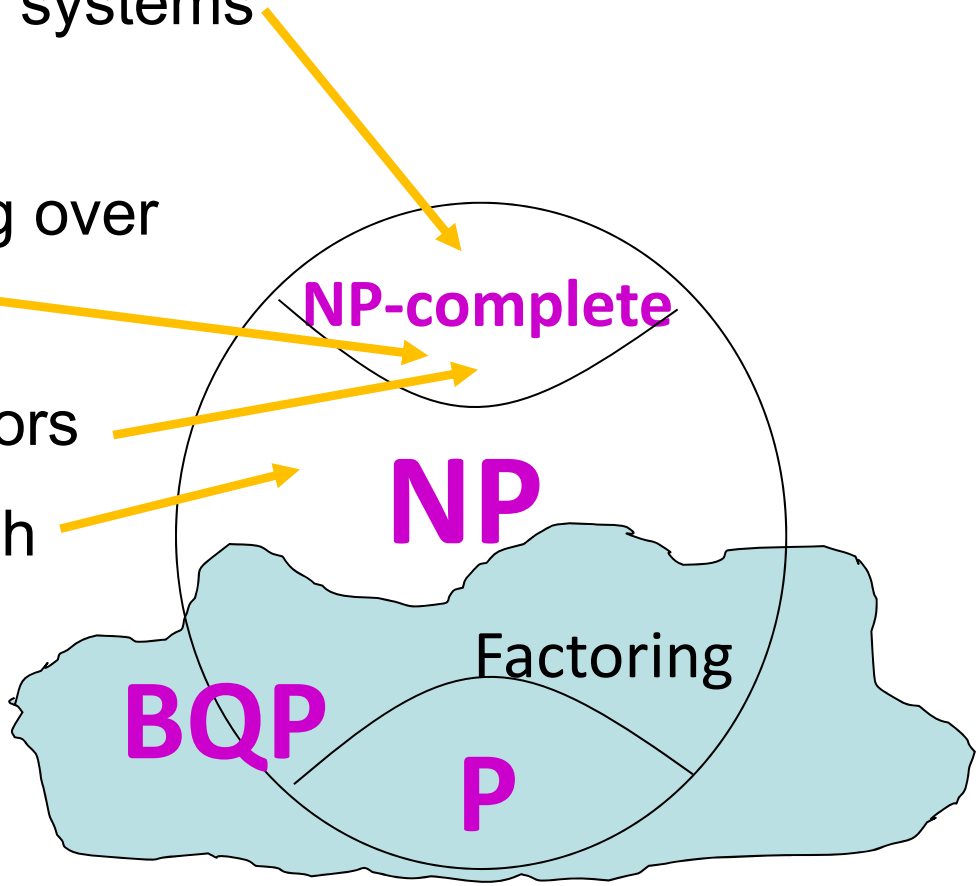


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Candidates

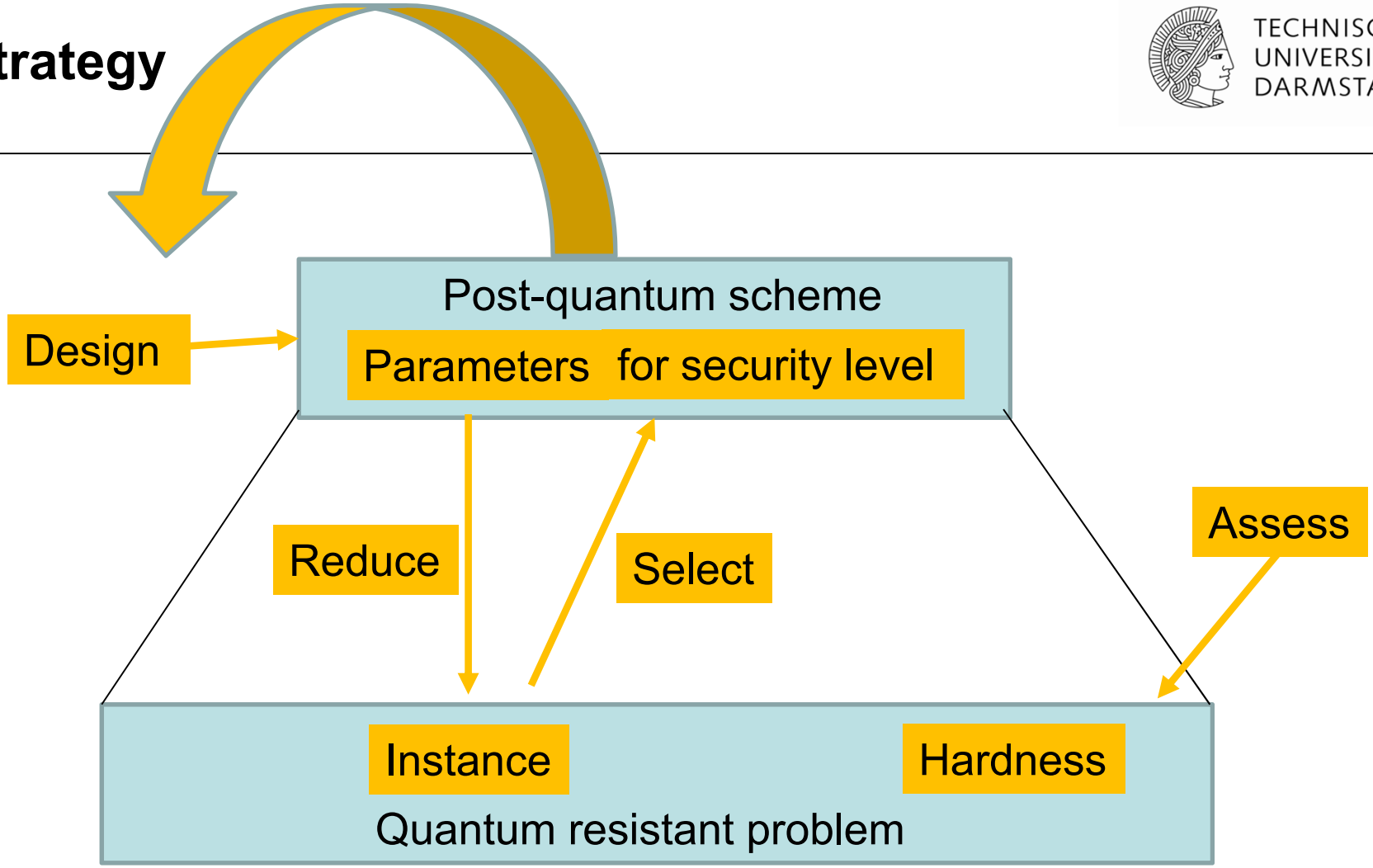
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- Solving non-linear equation systems over finite fields
- Bounded distance decoding over finite fields
- Short and close lattice vectors
- Breaking cryptographic hash functions



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Strategy


Post-quantum scheme
Parameters for security level
Design
Reduce
Select
Assess
Instance
Hardness
Quantum resistant problem

Strategy - Methodology

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Task	Goal	Method
Design	Efficient crypto with tight reduction proof to hard algorithmic problem	Algorithmics, quantum complexity theory
Assess	Quantum resistant problems, quantum time-space complexity, worst-to-average-case reduction	Quantum algorithmics, quantum complexity theory, parallel computing
Select	Parameter sets for given security level	Explicit complexity analysis

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Multivariate cryptography

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MQ problem

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$$4x + x^2 + y^2z \equiv 1 \mod 13$$

$$7y^2 + 2xz^2 \equiv 12 \mod 13$$

$$x + y^2 + 12xz^2 \equiv 4 \mod 13$$

Solution: $x = 15,\ y = 29,\ z = 45$

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MQ-Problem

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Given: $n, m, p_1, \ldots, p_m \in F[x_1, \ldots, x_n]$ quadratic, F finite field

Find: $y_1, \ldots, y_n \in F$, such that

$$p_1(y_1, \ldots, y_n) = \ \ldots = p_m(y_1, \ldots, y_n) = 0$$

MQ is NP-complete (Garey, Johnson 1979) (decision version)

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Fukuoka MQ Challenge
News
2015/08/04
Type III of n=34 and m=68 was solved by Tung Chou, Ruben Niederhagen, Bo-Yin Yang.
2015/07/13
Type IV of n=99 and m=66 was solved by Tung Chou, Ruben Niederhagen, Bo-Yin Yang.
2015/07/08
Type IV of n=97 and m=65 was solved by Tung Chou, Ruben Niederhagen, Bo-Yin Yang.
2015/07/01
Type IV of n=96 and m=64 was solved by Tung Chou, Ruben Niederhagen, Bo-Yin Yang.
Submission
Guide for Participants
How to participate
Challenge Format
Download Challenges
Encryption

Multivariate signatures

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$P: F^n \to F^m$, easily invertible non-linear

$S: F^n \to F^n$, $T: F^m \to F^m$, affine linear

Public key: $G = S \circ P \circ T$, hard to invert

Secret Key: S, P, T allows to find G^{-1}

$$G^{-1} = T^{-1} \circ P^{-1} \circ S^{-1}$$

Signing: $s = T^{-1} \circ P^{-1} \circ S^{-1}(m)$

Verifying: $G(s) =^{?} m$

Forging signature: Solve $G(s) - m = 0$

Fast

Large keys:
100 kBit for 100 bit security
Compared to
1776 bit
RSA modulus

- UOV , Goubin et al., 1999
- Rainbow, Ding, et al. 2005
- pFlash, Cheng, 2007
- Gui, Ding, Petzoldt, 2015

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Code-based cryptography

Bounded distance decoding problem

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Given:
- Linear code $C \subseteq F_2^n$
- $y \in F_2^n$
- $t \in \mathbb{N}$

Find:
- $x \in C$: $\mathrm{dist}(x, y) \leq t$

BDD is NP-complete (Berlekamp et al. 1978) (Decisional version)

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McEliece cryptosystem (1978)

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S, G, P matrices over F

G generator matrix for Goppa code ⟵ Allows to solve BDD

Public key: $G' = S \circ G \circ P,\ t$

Secret Key: P, S, G

Encryption: $c = mG' + z \in F^n$

Decryption: $x = cP^{-1} = mSG + zP^{-1}$

solve BDD to get $y = mSG$

decode to obtain m

Fast

Large public keys!
500 kBits for 100 bit security
Compared to 1776 bit RSA modulus

IND-CPA secure version

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Lattice-based cryptography

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Why lattice-based cryptography?

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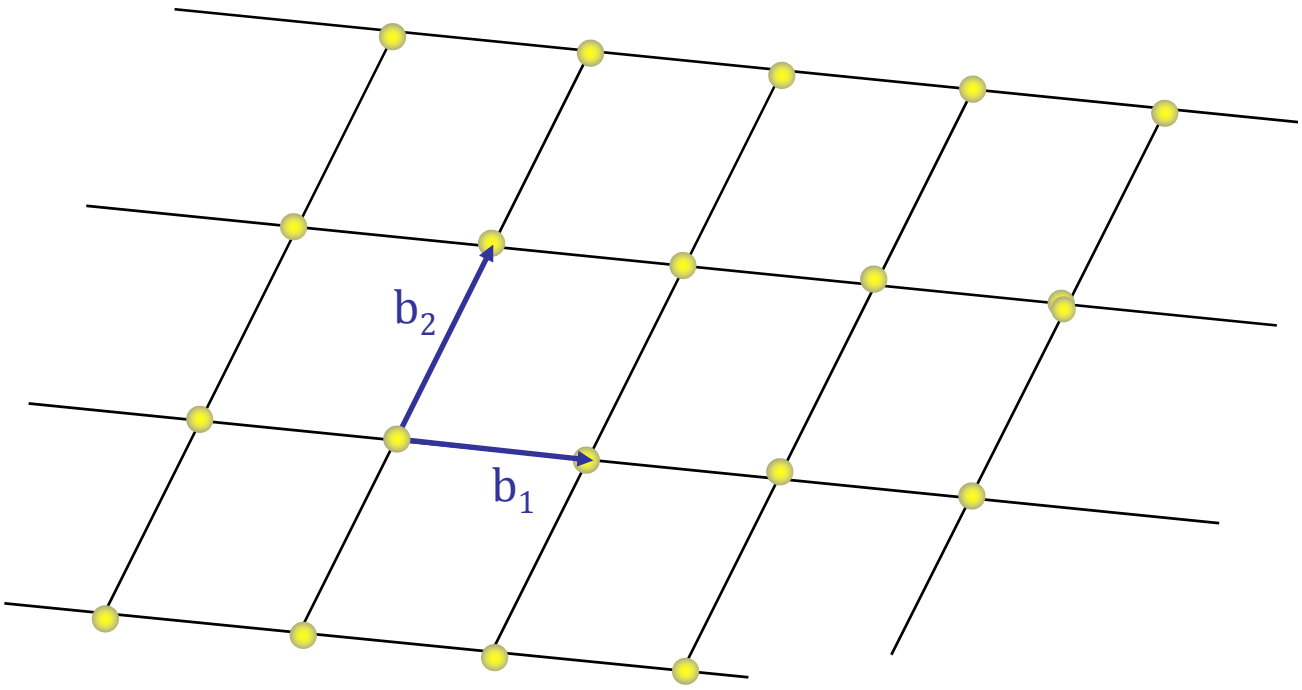
- Expected to resist quantum computer attacks
- Worst-to-average-case reduction
- Permits fully homomorphic encryption

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2-dimensional lattice

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$$B = (b_1, b_2), L(B) = \mathbb{Z}b_1 + \mathbb{Z}b_2$$



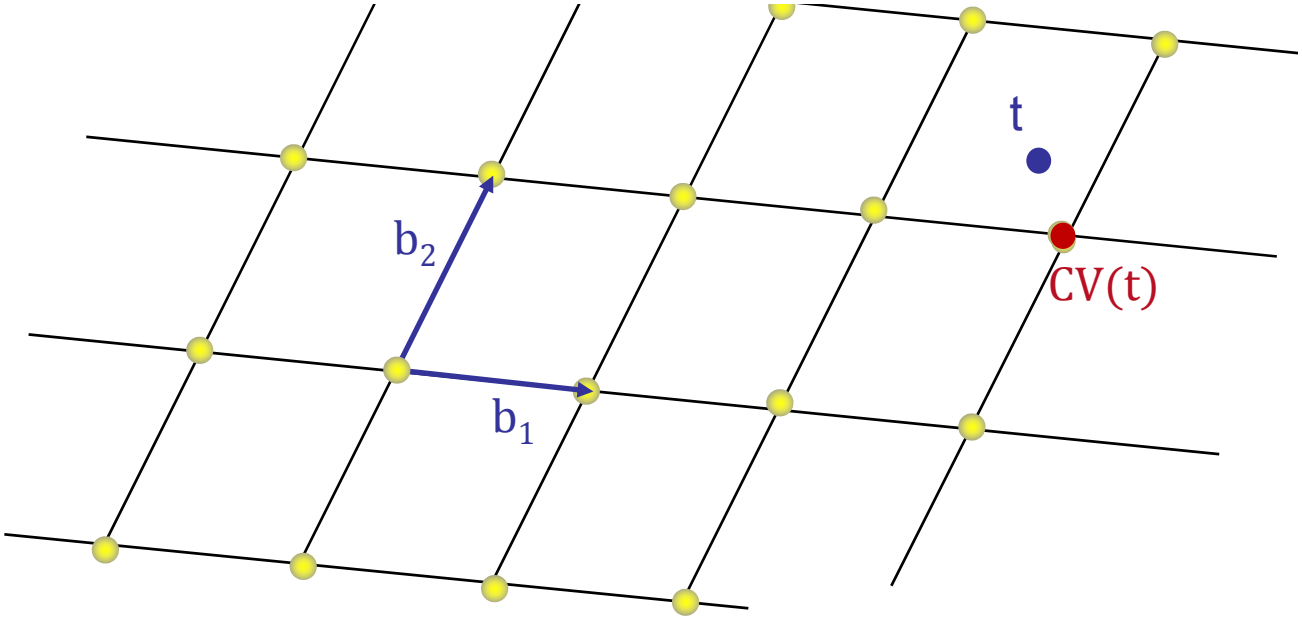
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2-dimensional αCVP

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Given: $B = (b_1, b_2),\ t, \alpha$

Find: $CV(t) \in L(B)$: $\left|\left|t - CV(t)\right|\right| \leq \alpha \min_{w \in L} ||t - w||$



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Lattice problems

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$n \in \mathbb{N}, L = \mathbb{Z}b_1 + \cdots + \mathbb{Z}b_n \subseteq \mathbb{R}^n$ lattice; B = (b_1, …, b_n) basis

α-Closest Vector Problem (CVP)

Given: $\alpha > 1$, lattice $L = L(B)$ basis B, t

Find: $v \in L$ such that $||t - v|| \leq \alpha \min_{w \in L} ||t - w||$

α-Shortest Vector Problem (SVP)

Given: $\alpha > 1$, lattice $L = L(B)$ in terms of basis B

Find: $v \in L$ nonzero such that $||v|| \leq \alpha \lambda_1(L)$

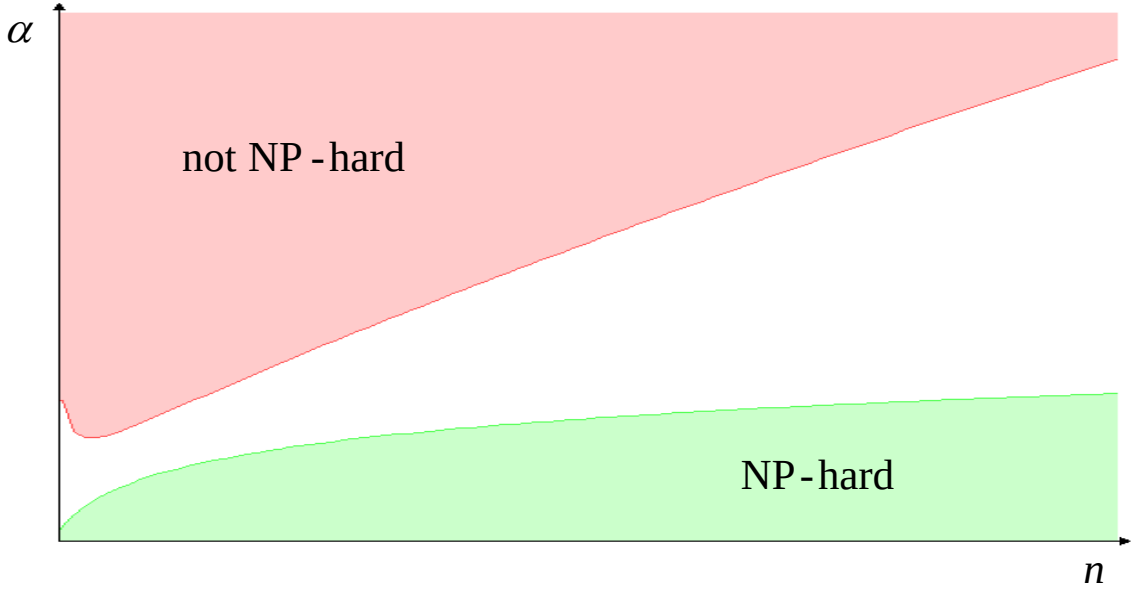
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Complexity of α-CVP/SVP

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Arora et al. (1997):

$$\log(n)^c \text{-CVP/SVP is NP-hard for all } c$$



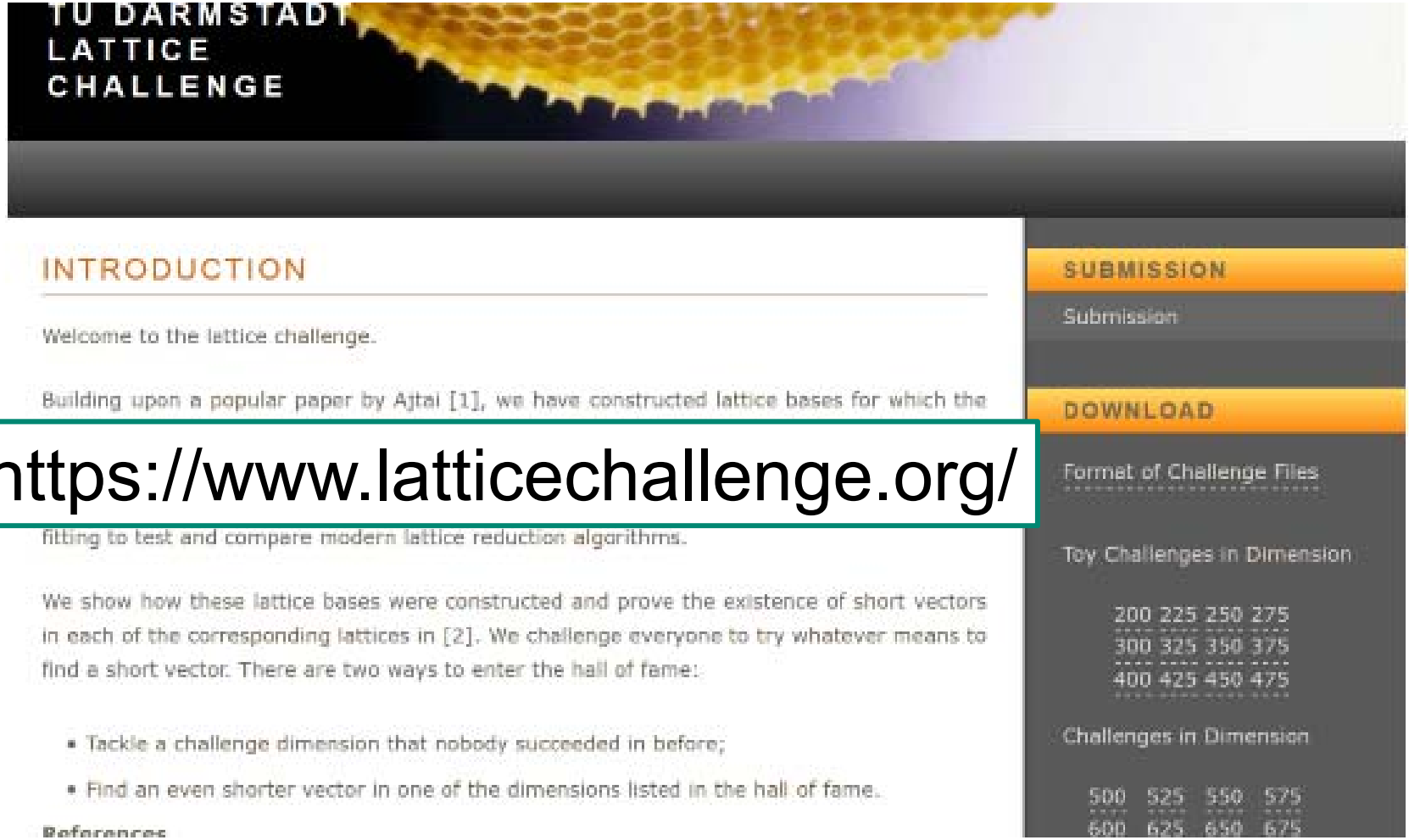
Goldreich, Goldwasser (2000):

$$\Omega\left(\sqrt{n}/\log(n)\right)\text{-CVP/SVP is not NP-hard or } \mathbf{coNP} \subseteq \mathbf{AM}$$

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Practical complexity

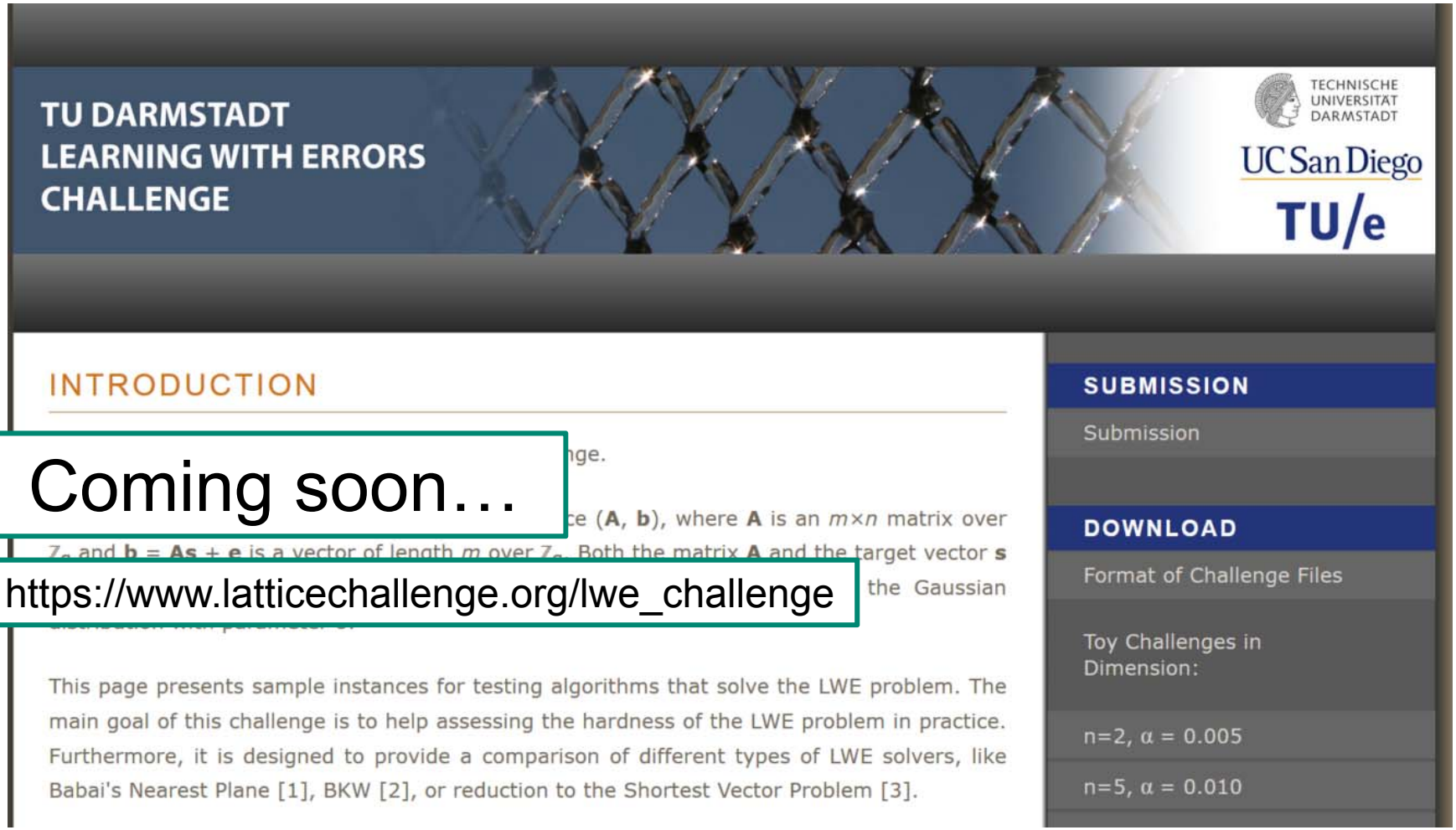
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Practical complexity

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Coming soon…

https://www.latticechallenge.org/lwe_challenge

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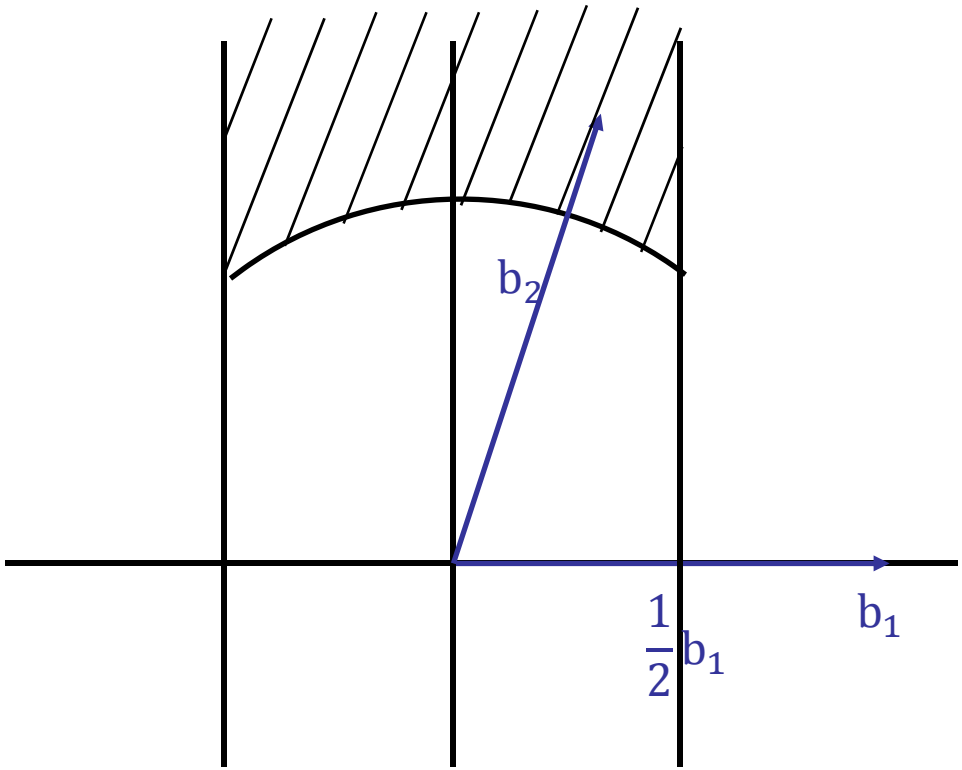
The idea of lattice-based cryptography

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- **GGH Sign 1995**
- NTRU Encrypt 1996
- NTRU Sign 2003

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Reduced bases (Gauß 1801)

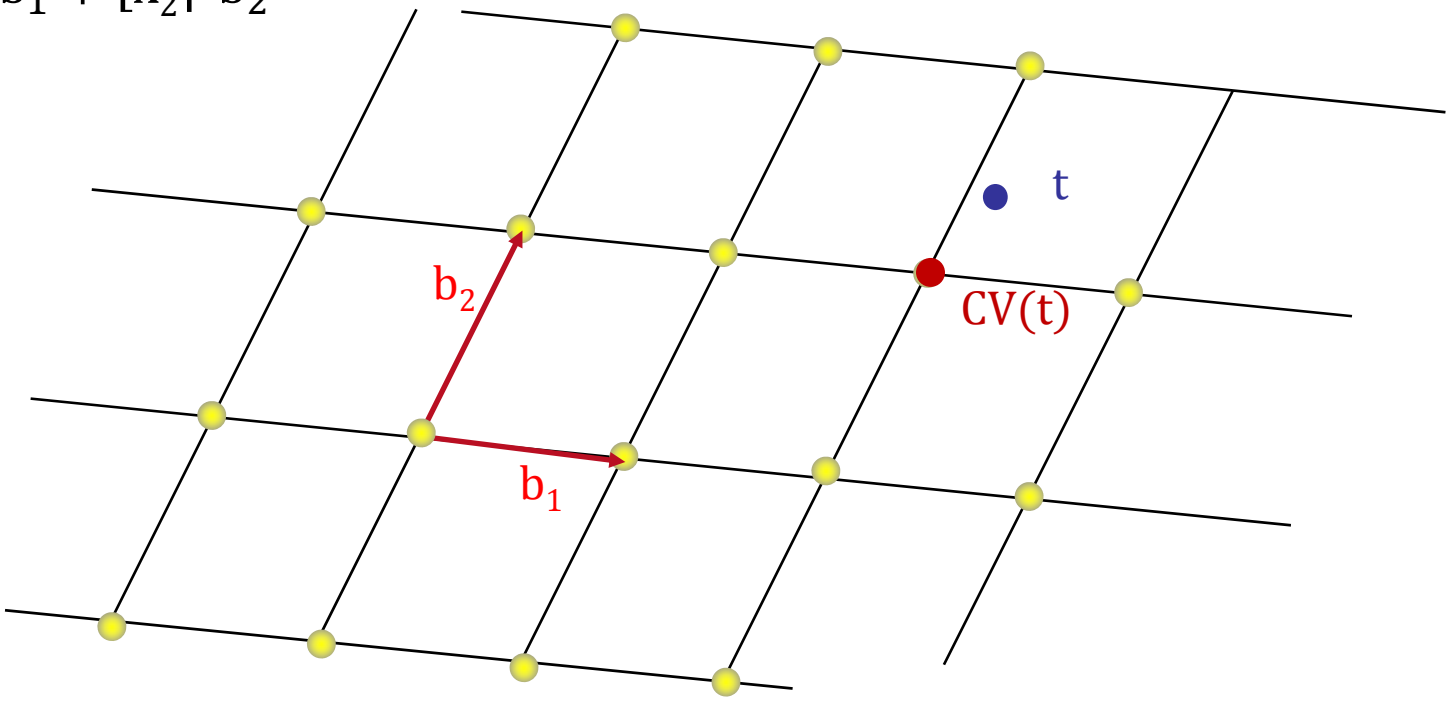


$(\mathbf{b_1}, \mathbf{b_2})$ reduced
Gauss: (SVP and) CVP easy

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$$t = x_1 b_1 + x_2 b_2$$

$$CV(t) = \lfloor x_1 \rceil b_1 + \lfloor x_2 \rceil b_2$$



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$\mathbf{B} = (\mathbf{b_1}, \mathbf{b_2})$ not reduced ⇒ CVP hard

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$L = \mathbb{Z}^2$, $B = \left(\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right)$, $t = \begin{pmatrix}3.4\\-2.3\end{pmatrix}$, $CVP(t) = \begin{pmatrix}3\\-2\end{pmatrix}$

Another basis $B' = \left(\begin{pmatrix}100\\99\end{pmatrix}, \begin{pmatrix}99\\98\end{pmatrix}\right)$

$$t = \begin{pmatrix}3.4\\-2.3\end{pmatrix} = -560.9 \cdot \begin{pmatrix}100\\99\end{pmatrix} + 566.6 \cdot \begin{pmatrix}99\\98\end{pmatrix}$$

$$-561 \cdot \begin{pmatrix}100\\99\end{pmatrix} + 567 \cdot \begin{pmatrix}99\\98\end{pmatrix} = \begin{pmatrix}33\\27\end{pmatrix} \neq \begin{pmatrix}3\\-2\end{pmatrix} = CVP(t)$$

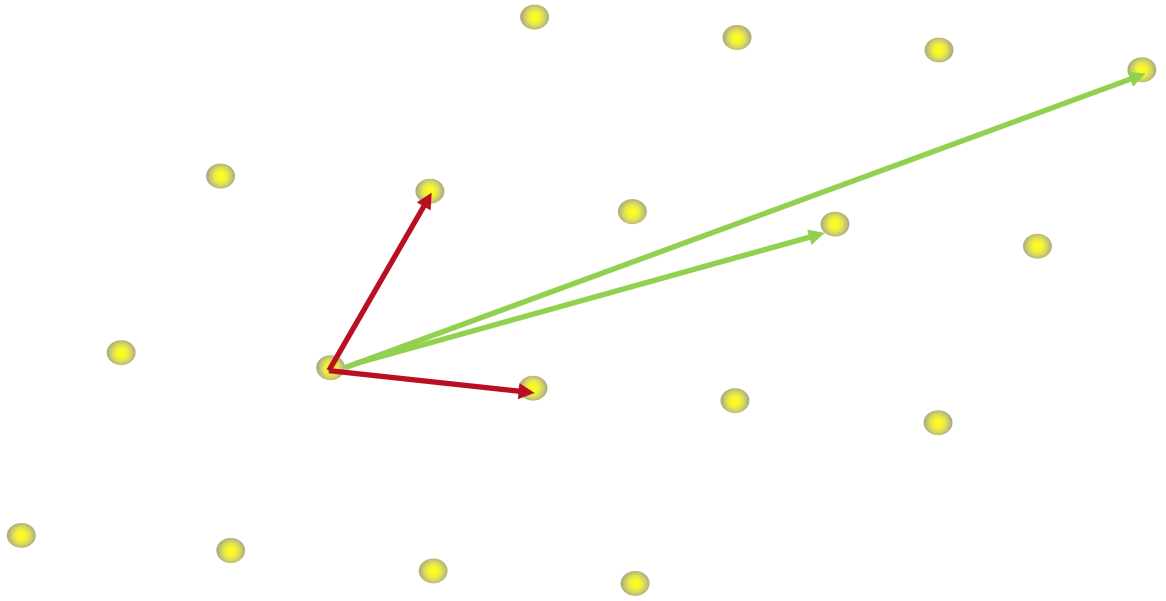
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Key generation

Key generation: $n \in \mathbb{N}, L \subseteq \mathbb{R}^n$ lattice

Secret key: „reduced“ basis B of L. (Allows to efficiently solve CVP.)

Public key: „bad” basis B' of L. (Does not.)



Public-key encryption

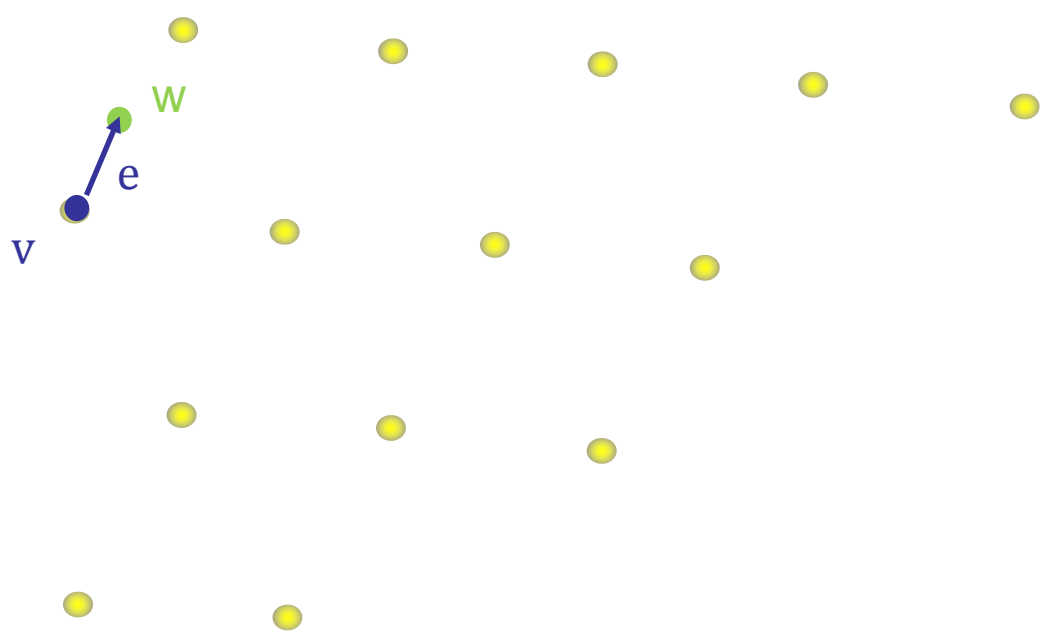
Plaintext $v \in L$

Encryption(public key, v)

- small $e \in \mathbb{R}^n$
- ciphertext $w = v + e$

Decryption(secret key, w):

- $v = CV(w)$



Digital signature

Public: Cryptographic hash function $h: \{0,1\} \rightarrow \mathbb{R}^n$

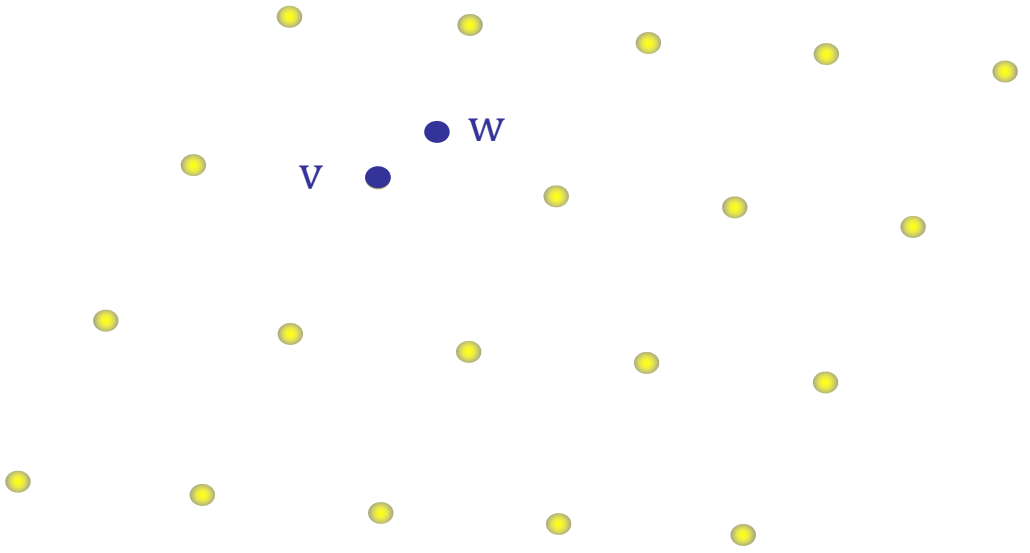
Sign(secret key, document d):

$$w = h(d)$$
$$v = CV(w)$$

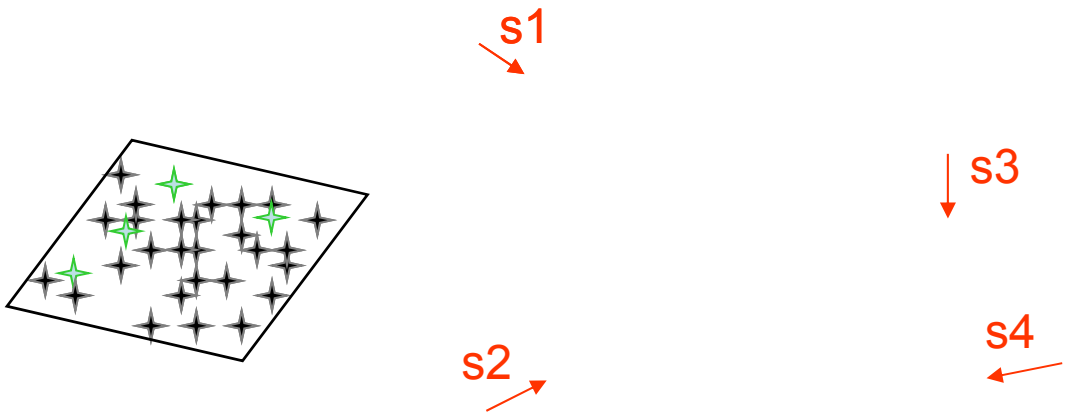
Verify(public key, v, w):

v lattice vector?

v close to w ?



Learning the secret key
Nguyen and Regev 2006



NTRU-251 broken using ≈ 400 signatures

GGH-400 broken using ≈ 160.000 signatures

TESLA Signature Scheme – Performance

Bindel et al.

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	Sign (ms)	Verify (ms)	Signature (bit)	Public Key (byte)	Secret Key (byte)	Bit Security
TESLA-128	0,19	0,07	10,240	1,331,200	1,011,712	128
RSA 4096	8,66	0,14	4,096	512	512	128

Encryption Scheme - Performance

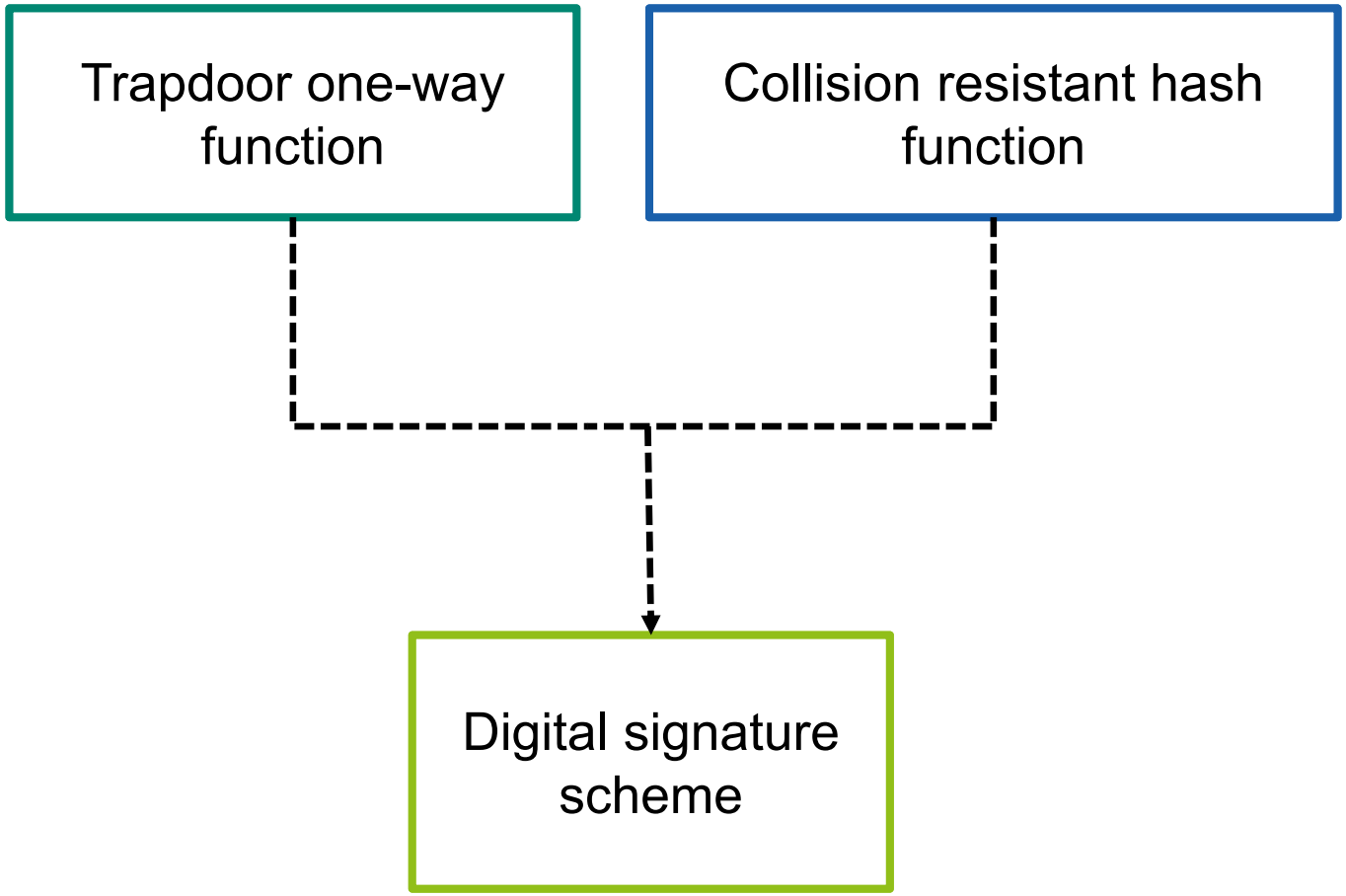
El Bansarkhani et al.

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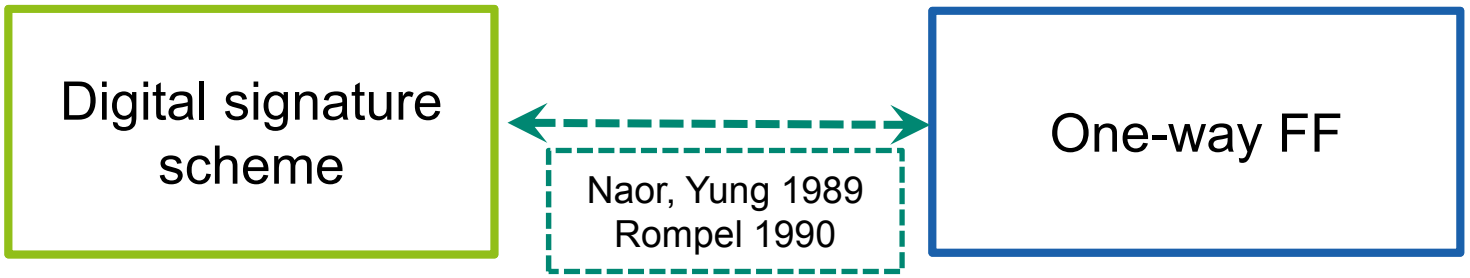
	Encrypt ms	Decrypt ms	Ciphertext (bit)	Public Key (byte)	Secret Key (byte)	Expansion Factor	Bit Security
Our Scheme	0.07	0.05	9,216	1,472	768	3.8	100
Our Scheme	0.22	0.09	46,080	1,472	768	2.3	100
RSA 2048	2.4 ms	4.3	2,048	256	256	1	100

Hash-based signatures

Typical construction



Trapdoor one-way functions hard to construct but not required

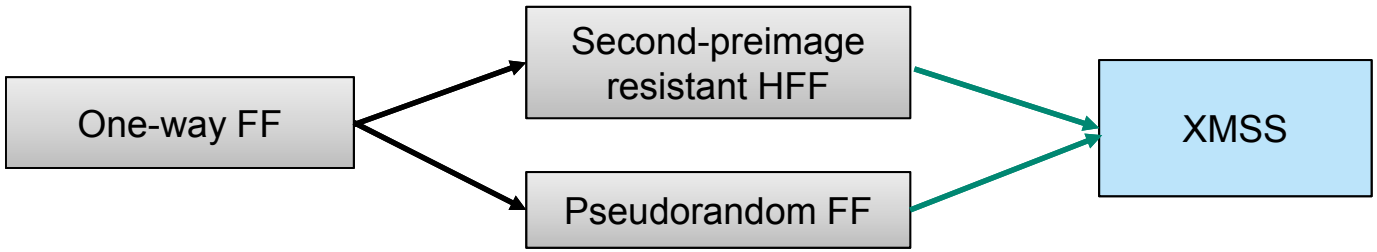


XMSS signature

JB, Coronado, Dahmen, Hülsing

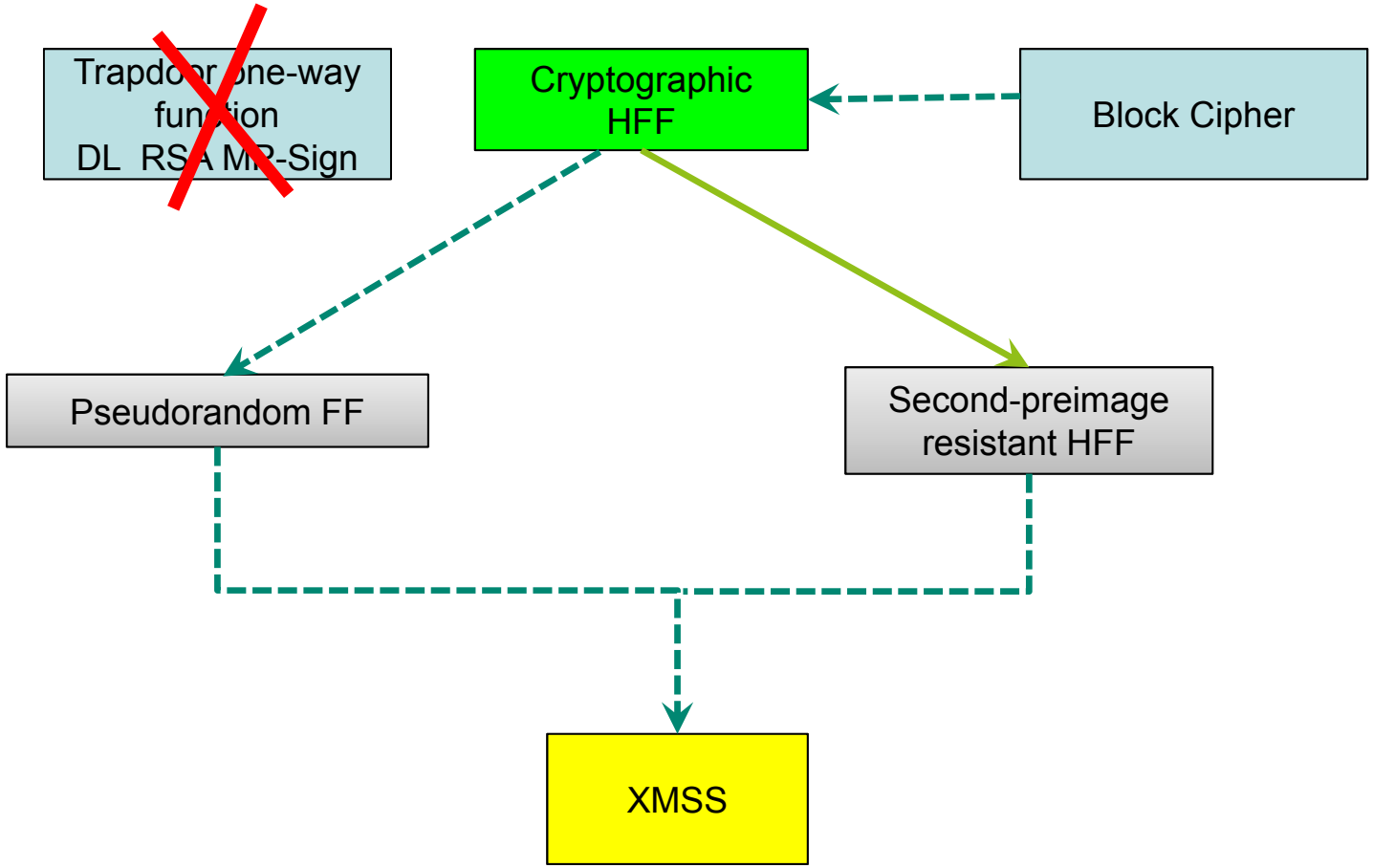
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- Based on Merkle signature scheme
- Has minimal security requirements



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XMSS in practice



Hash functions & Blockciphers

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AES
Blowfish
3DES
Twofish
Threefish
Serpent
IDEA
RC5
RC6
…

SHA-2
SHA-3
BLAKE
Grøstl
JH
Keccak
Skein
VSH
MCH
MSCQ
SWIFFTX
RFSB
…

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XMSS performance

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	Sign (ms)	Verify (ms)	Signature (bit)	Public Key (bit)	Secret Key (byte)	Bit Security	Comment
XMSS-SHA-2	35.60	1.98	**16,672**	13,600	3,364	157	h = 20, w = 64,
XMSS-AES-NI	**0.52**	**0.07**	19,616	7,328	1,684	84	h = 20, w = 4
XMSS-AES	1.06	0.11	19,616	7,328	1,684	84	h = 20, w = 4
RSA 2048	**3.08**	**0.09**	≤ 2,048	≤ 4,096	≤ 512	87	

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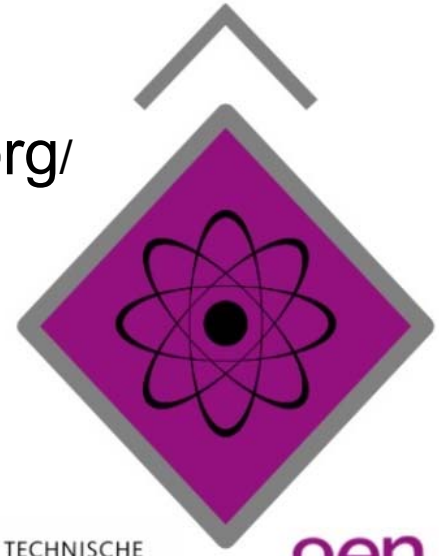
XMSS transfer project

Denis Butin, Stefan Gazdag

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Practical Hash-based Signatures

http://www.square-up.org/



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genua

Soviel ist sicher.

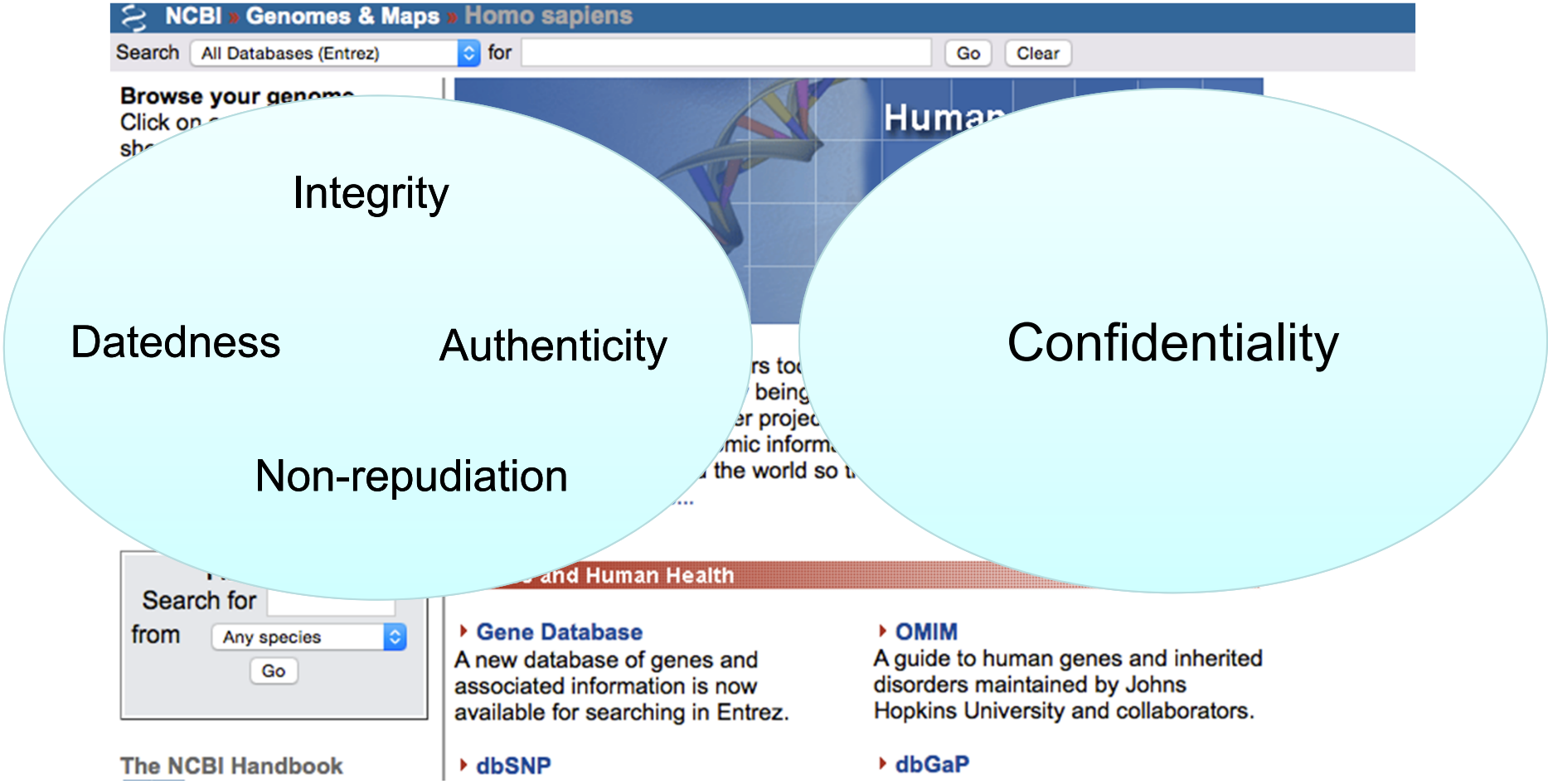
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Quantum cryptography?

Source of randomness!

Long-lived systems

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Long-term confidentiality

Braun, Buchmann, Mullan, Wiesmaier
“Long term confidentiality: a survey”
Designs, Codes and Cryptography, 2012

Computational security sufficient?

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Secure until	Symmetric	RSA	GFP	ECC
2015	80	1248	1248	160
2020	96	1776	1776	192
2030	112	2432	2432	224
2040	128	3248	3248	256
future	256	15424	15424	512

Ecrypt II recommendations 2012



Store ciphertexts now – decrypt them later

Computational security not sufficient!

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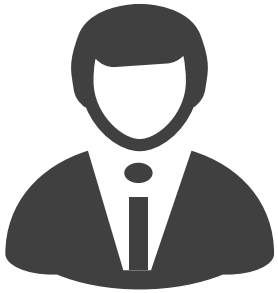
NSA data center Utah – $3x10^{18}$ - 10^{24} bytes



Store ciphertexts now – decrypt later

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Long-term confidentiality





Data **in transit**



Data **at rest**

Long-term confidentiality of **data in transit**

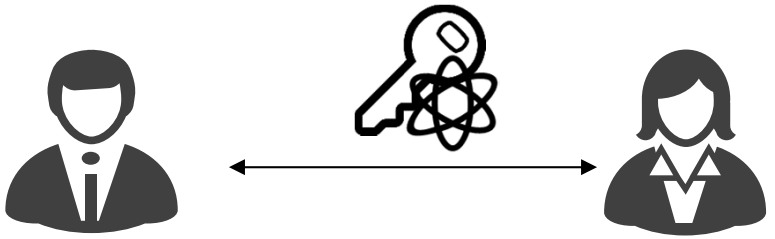
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Shannon: 1949: Information theoretic secure one-time pad encryption



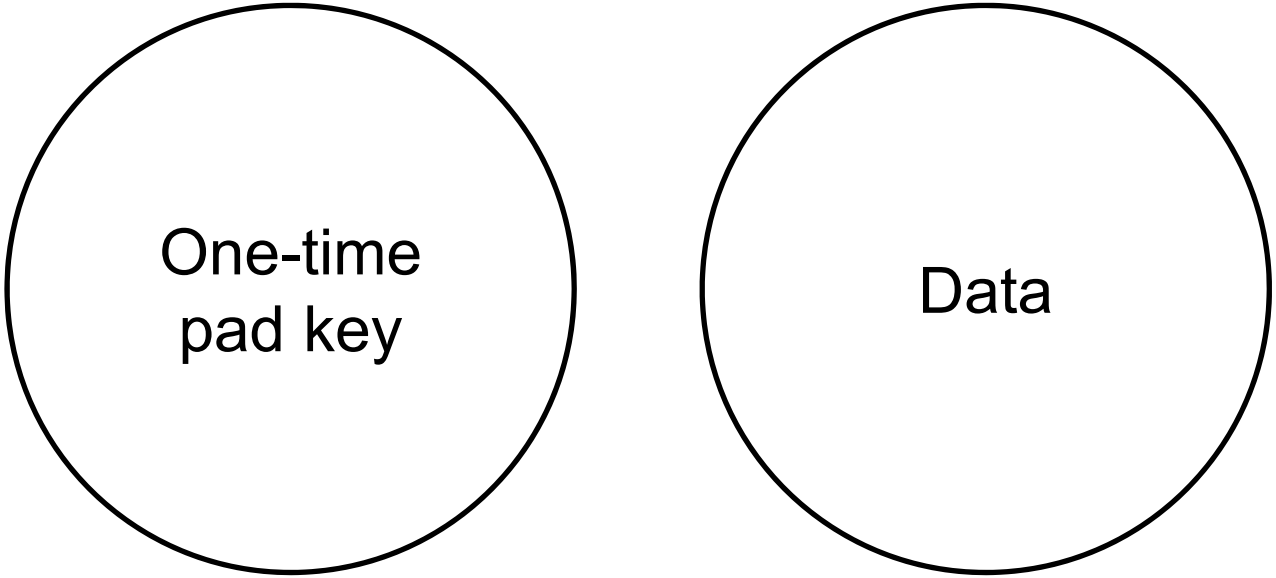
Key distribution?

BB84: Quantum key distribution



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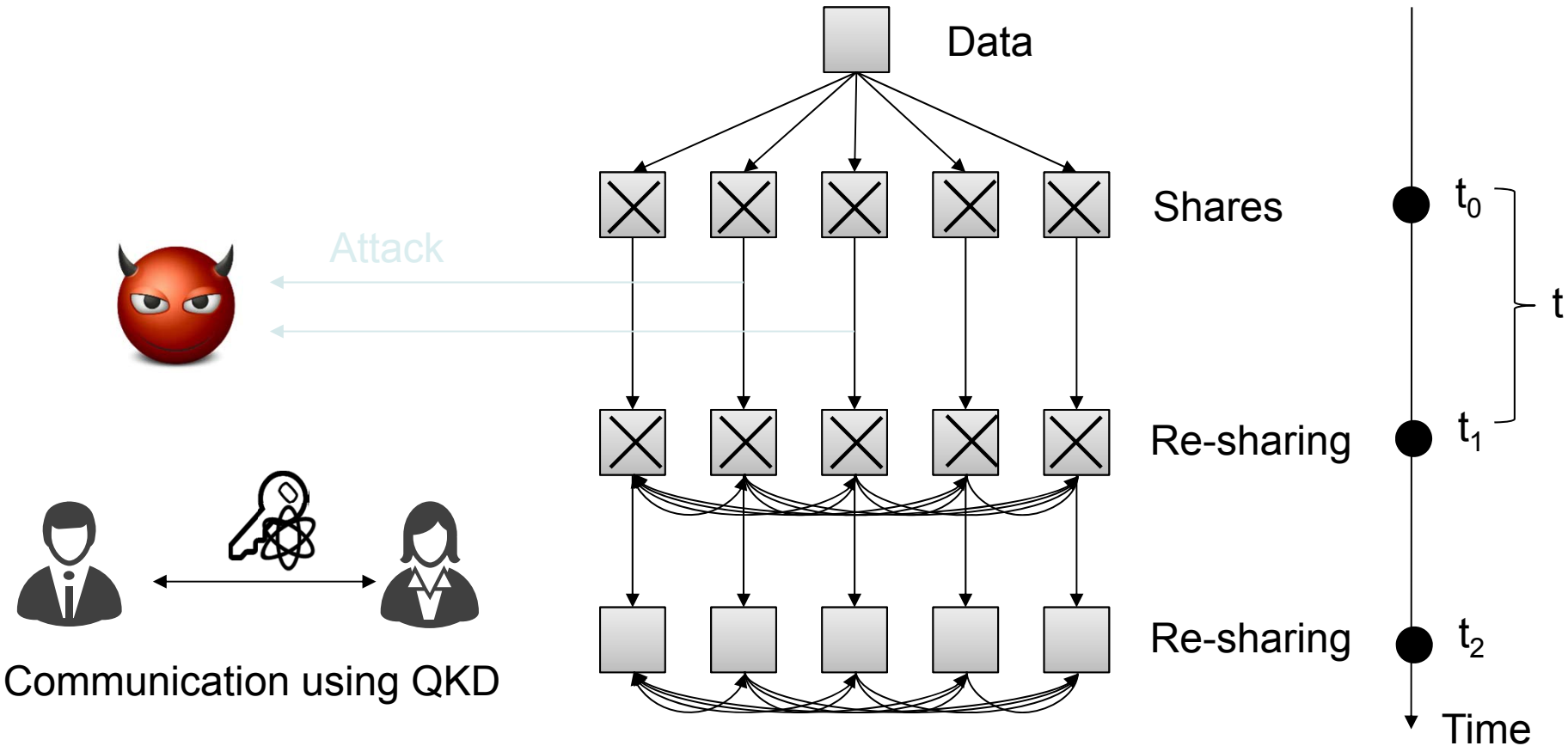
Long-term confidentiality of **data at rest**



Key size = Message size

➔ One-time pad useless

Proactive secret sharing



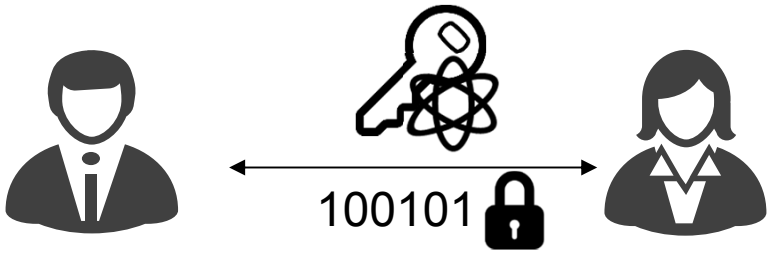
Security assumption: adversary unable to access ≥3 shares **over time period t**

Long-term confidentiality – **challenges**

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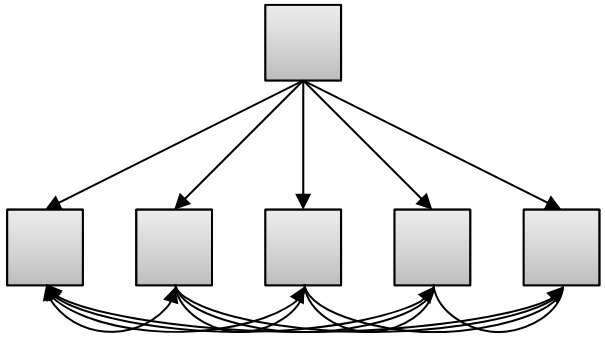
Data in transit

- Integrate QKD with secure channel protocol



Data at rest

- Reduce communication complexity



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Conclusion

Todos

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- Standardize and integrate into standard applications: XMSS + NTRU-Encrypt/McEliece

- Provide/optimize security proofs



http://www.crossing.tu-darmstadt.de

- Study computational problems in the presence of modern computing architectures -> parameter selection

- Optimize schemes for secure parameters - consider side channels.

- Integrate with quantum cryptography.

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PQCrypto 2016

About the Conference

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Call for Papers

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Registration

Program

The Seventh International Conference on Post-Quantum Cryptography

Fukuoka, Japan, February 24-26, 2016



7 submission deadline is fast approaching!

, 2015: The winter school lecturers are announced.

15: The submission server is now open.

e Conference

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http://www.crossing.tu-darmstadt.de

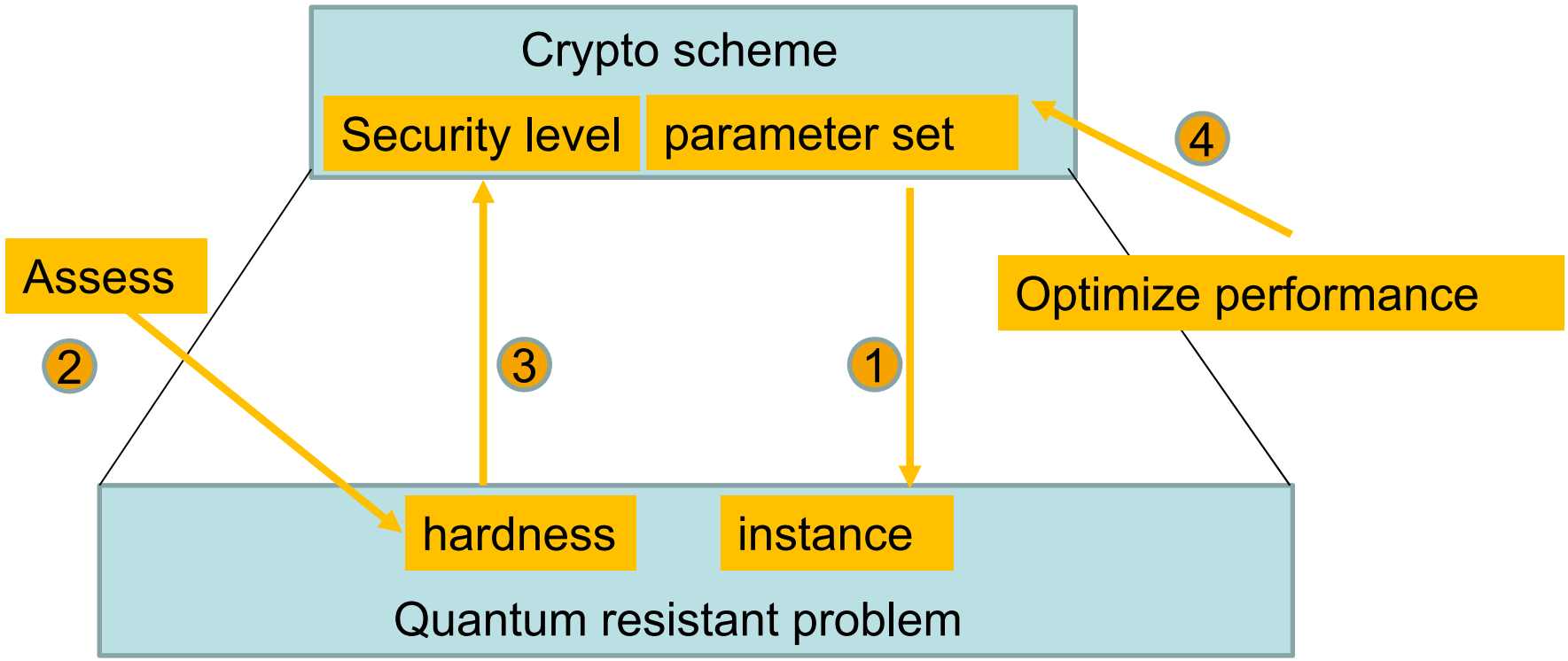
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Strategy

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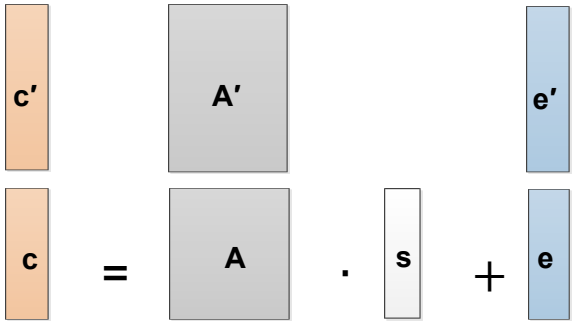
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Folie von Rachid

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Ingrediens Encryption Scheme,

- Ciphertext is A-LWE instance **c**
- Error-term **e** contains the message
- Flexible increase of message size via **e′** and **c′**
- Uniform random **e** possible
  → Faster algorithms, msg exp. factors ≈ 1
- Decryption requires trapdoor for **A**
- Further features: **e′** can be a signature, etc.



	Key sizes (bytes)	#Msg bits (encrypted)	Expansion Factor	Encrypt Cycles/bit	Decrypt Cycles/bit	Bit Security
Our Scheme	pk: 1472 sk: 768	9216	3.8	13.1 [0.07 ms]	10.7 [0.05 ms]	100
	pk: 1472 sk: 768	46080	2.3	8.4 [0.22 ms]	3.6 [0.09 ms]	100
RSA	pk: 256 sk: 256	2048	1	2134 [2.4 ms]	3756 [4.3 ms]	100

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Performance

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- NTRU encrypt 1996: fast and small

The provable schemes to be studied more

- BLISS 2013 and Bai-Galbraith 2014 signature with improvements of Bindel: fast but large key size

- Lindner, Peikert 2011 encryption with improvements of Göpfert: fast but ciphertext expansion

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